

$$27 | 3 | 17$$

$$98 + 3 - 9 = 92 \quad (1+2+5+4=12m)$$

DETERMINANTS

Determinant of a 3rd order matrix :

$$\text{Let } A = \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix}$$

along R_1 :

$$\Delta A = a_1(b_2c_3 - b_3c_2) - a_2(b_1c_3 - b_3c_1) + a_3(b_1c_2 - b_2c_1)$$

$$\text{Minor of } b_1 = \begin{vmatrix} a_2 & a_3 \\ c_2 & c_3 \end{vmatrix}$$

$$\text{Cofactor of } b_1 = - \begin{vmatrix} a_2 & a_3 \\ c_2 & c_3 \end{vmatrix}$$

$$\Delta A = \text{along } C_3 = a_3 \begin{vmatrix} b_1 & b_2 \\ c_1 & c_2 \end{vmatrix} - b_3 \begin{vmatrix} a_1 & a_2 \\ c_1 & c_2 \end{vmatrix} + c_3 \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix}$$

$$\text{Eg: } \Delta = \begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix}$$

$$\Delta_1 = 1(45 - 48) - 2(36 - 42) + 3(32 - 35)$$

$$= -3 + 12 - 9 = 0$$

$$\Delta_2 = 3(32 - 35) - 6(8 - 14) + 9(5 - 8)$$

$$= -9 + 36 - 27 = 0$$

EXERCISE 4.1

$$1] \begin{vmatrix} 2 & 4 \\ -5 & -1 \end{vmatrix} = -2 + 20 = 18$$

$$2] (i) \begin{vmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{vmatrix} = \cos^2 \theta + \sin^2 \theta = 1$$

$$\begin{aligned} \text{(ii)} \quad \begin{vmatrix} x^2-x+1 & x-1 \\ x+1 & x+1 \end{vmatrix} &= (x^2-x+1)(x+1) - (x+1)(x-1) \\ &= x^3-x^2+x+x^2-x+1 - x^2+1 \\ &= x^3-x^2+2 \end{aligned}$$

$$5) \text{ (i)} \quad \begin{vmatrix} 3 & -1 & -2 \\ 0 & 0 & -1 \\ 3 & -5 & 0 \end{vmatrix}$$

$$\begin{aligned} \Delta &= 3(0-5) + 1(0+3) - 2(0-0) \\ &= -15 + 3 - 0 \\ &= -12 \end{aligned}$$

$$\text{(ii)} \quad \begin{vmatrix} 3 & -4 & 5 \\ 1 & 1 & -2 \\ 2 & 1 & 3 \end{vmatrix} \quad \text{column:}$$

$$\begin{aligned} \Delta &= 3(1+6) + 4(1+4) + 5(3-2) \\ &= 21 + 20 + 5 \\ &= 46 \end{aligned}$$

$$\Delta_1 = \begin{vmatrix} 1 & -4 & 5 \\ 2 & 1 & -2 \\ 3 & 3 & 1 \end{vmatrix}$$

$$\begin{aligned} \Delta &= 1(1+6) + 4(2+6) + 5(6-3) \\ &= 7 + 32 + 15 \\ &= 54 \end{aligned}$$

$$\begin{aligned} \Delta_2 &= \begin{vmatrix} 3 & 1 & 5 \\ 2 & 3 & 1 \end{vmatrix} = 3(2+6) - 1(1+4) + 5(3-4) \\ &= 24 - 5 + 35 = 54 \end{aligned}$$

$$\begin{aligned} \Delta_3 &= \begin{vmatrix} 3 & -4 & 1 \\ 1 & 1 & 2 \\ 2 & 3 & 3 \end{vmatrix} = 3(3+6) + 4(3+4) + 1(3-2) \\ &= 27 + 28 + 1 = 56 \end{aligned}$$

Achiever

(*) $\Delta = \begin{vmatrix} 0 & 1 & 2 \\ 1 & -2 & 3 \\ 1 & 0 & -4 \end{vmatrix} = \begin{vmatrix} 1 & 2 \\ 0 & 1 \end{vmatrix} = 1 \times 1 - 0 \times 2 = 1$

$\Delta = 0(8-0) - 1(4-3) + 2(0+2)$
 $= 0 + 7 + 4$
 $= 11$

$\Delta_1 = \begin{vmatrix} -1 & 2 & 0 \\ 0 & -2 & 3 \\ 1 & 0 & -4 \end{vmatrix} = -1(-8-0) - 1(0-3) + 2(0+2)$
 $= -8 + 3 + 4 = -1$

$\Delta_2 = \begin{vmatrix} 0 & -1 & 2 \\ 1 & 0 & 3 \\ 1 & 1 & -4 \end{vmatrix} = 0(0-3) + 1(4-3) + 2(1-0)$
 $= 0 + 1 + 2 = 3$

$\Delta_3 = \begin{vmatrix} 0 & 1 & -1 \\ 1 & -2 & 0 \\ 1 & 0 & 1 \end{vmatrix} = 0(-2-0) - 1(1-0) - 1(0+2)$
 $= 0 - 1 - 2 = -3$

(iii) $\Delta = \begin{vmatrix} 0 & 1 & 2 \\ -1 & 0 & -3 \\ -2 & 3 & 0 \end{vmatrix} = 0(0-6) - 1(-3+6) + 2(-3+0)$
 $= 0 + 3 - 6 = -3$

$\Delta = \begin{vmatrix} 2 & -1 & -2 \\ 0 & 2 & -1 \\ 3 & -5 & 0 \end{vmatrix} = 2(0-5) + 1(0+3) - 2(0-6)$
 $= -10 + 3 + 12 = 5$

3] If $A = \begin{bmatrix} 1 & 2 \\ 4 & 2 \end{bmatrix}$ show that $|2A| = 4|A|$

LHS: $|2A| = \begin{vmatrix} 2 & 4 \\ 8 & 4 \end{vmatrix} = 8 - 32 = -24$

RHS: $4|A| = 4(2-8) = 4(-6) = -24$

LHS = RHS

4] If $A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 4 \end{bmatrix}$ show that $|3A| = 27|A|$

$$\text{LHS: } |3A| = \begin{vmatrix} 3 & 0 & 3 \\ 0 & 3 & 6 \\ 0 & 0 & 12 \end{vmatrix} = 3(36 - 0) - 0(0 - 0) + 3(0 - 0) = 108$$

$$\text{RHS: } 27|A| = 27[1(4 - 0) - 0(0 - 0) + 1(0 - 0)] = 27(4) = 108$$

$$\text{LHS} = \text{RHS}$$

$$|kA|_{n \times n} = k^n |A|$$

Eg: $A = \begin{bmatrix} 1 & 2 \\ 4 & 2 \end{bmatrix}$ (i) find $|2A|$

$$|2A| = 2^2 |A| = 4|A| = 4(2 - 8) = -24$$

(ii) find $|3A| = 3^2 |A| = 9(-6) = -54$

Eg: $A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 4 \end{bmatrix}_{3 \times 3}$ find $|4A|$

$$|4A| = 4^3 |A| = 64[1(4)] = 256$$

Eg: If A is a matrix of order 4 & $|A| = -3$. Find $|2A|$.

Sol: Given: order = 4, $|A| = -3$

$$|2A| = 2^4 |A| = 16(-3) = -48$$

Eg: If A is a matrix of order 3 & $|A| = 4$, Find $|5A|$.

Sol: $|5A| = 5^3 (4) = 125(4) = 500$

Solving equations by matrix method :- $A \cdot X = P$

- 1] matrix of cofactors : is the matrix which gives the cofactors of every element of any square matrix A .
- 2] Minor of a_{ij} : minor of an element a_{ij} is obtained by taking the determinant after deleting i^{th} row and j^{th} column and is denoted as M_{ij} .
- 3] Cofactor of a_{ij} : cofactor of an element a_{ij} is defined as $(-1)^{i+j} \times M_{ij}$ and is denoted as A_{ij} .

$$\therefore A_{ij} = (-1)^{i+j} M_{ij}$$

Sign Convention in a matrix :

Every element in a matrix has a sign given by

$(-1)^{i+j}$ if

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} (-1)^{1+1} & (-1)^{1+2} & (-1)^{1+3} \\ (-1)^{2+1} & (-1)^{2+2} & (-1)^{2+3} \\ (-1)^{3+1} & (-1)^{3+2} & (-1)^{3+3} \end{bmatrix}$$

$$\therefore \begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$$

Adjoint of a matrix $A = \text{adj} A = \text{Transpose of matrix of cofactors}$.

i.e, $\text{adj} A = C'_A$

Eg: $A = \begin{bmatrix} 2 & -3 & 5 \\ 6 & 0 & 4 \\ 1 & 5 & -7 \end{bmatrix}$

$$C_A = \begin{bmatrix} 0 \cdot 4 & 6 \cdot 4 & 6 \cdot 0 \\ -3 \cdot 5 & 2 \cdot 5 & 2 \cdot -3 \\ 5 \cdot -7 & 1 \cdot -7 & 1 \cdot 5 \end{bmatrix} = \begin{bmatrix} -20 & 46 & 30 \\ -15 & 10 & -6 \\ -35 & -7 & 5 \end{bmatrix}$$

$$|A| = -40 - 3(46) + 150 = -28$$

$$\text{adj } A = C_A^T = \begin{bmatrix} -20 & 46 & 30 \\ -15 & 10 & -6 \\ -35 & -7 & 5 \end{bmatrix}$$

EXERCISE 4.5

Find adj of the following:

1] $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ $\text{adj } A = C_A^T = \begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix}$

2] $A = \begin{bmatrix} 1 & -1 & 2 \\ 2 & 3 & 5 \\ -2 & 0 & 1 \end{bmatrix}$

$$C_A = \begin{bmatrix} 3 \cdot 5 & 2 \cdot 5 & 2 \cdot 3 \\ -1 \cdot 2 & 1 \cdot 2 & 1 \cdot -1 \\ 0 \cdot 1 & -2 \cdot 1 & -2 \cdot 0 \end{bmatrix} = \begin{bmatrix} 15 & 10 & 6 \\ -2 & 2 & -1 \\ 0 & -2 & 0 \end{bmatrix}$$

$$|A| = 3 + 12 + 12 = 27$$

$$\text{adj } A = C_A^T = \begin{bmatrix} 15 & -2 & 0 \\ 10 & 2 & -2 \\ 6 & -1 & 0 \end{bmatrix}$$

Achiever

Verify $A \cdot (\text{adj } A) = (\text{adj } A) \cdot A = |A| I$

$$3] A = \begin{bmatrix} 2 & 3 \\ -4 & -6 \end{bmatrix} \quad |A| = -12 + 12 = 0$$

$$\text{adj } A = \begin{bmatrix} -6 & -3 \\ 4 & 2 \end{bmatrix}$$

$$\text{LHS} = A(\text{adj } A) = \begin{bmatrix} 2 & 3 \\ -4 & -6 \end{bmatrix} \begin{bmatrix} -6 & -3 \\ 4 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\text{RHS} = |A| I = 0 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\text{LHS} = \text{RHS}$$

$$4] A = \begin{bmatrix} 1 & -1 & 2 \\ 3 & 0 & -2 \\ 1 & 0 & 3 \end{bmatrix}$$

$$C_A = \begin{bmatrix} 0 & -2 & 3-2 & 3 & 0 \\ 0 & 3 & 1 & 3 & 1 & 0 \\ -1 & 2 & 1 & 2 & 1 & -1 \\ 0 & 3 & 1 & 3 & 1 & 0 \\ -1 & 2 & 1 & 2 & 1 & -1 \\ 0 & -2 & 3 & -2 & 3 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -11 & 0 \\ 3 & 1 & -1 \\ 2 & 8 & 3 \end{bmatrix}$$

$$|A| = 0 + 11 + 0 = 11; \quad \text{adj } A = C_A' = \begin{bmatrix} 0 & 3 & 2 \\ -11 & 1 & 8 \\ 0 & -1 & 3 \end{bmatrix}$$

$$\text{LHS} = A(\text{adj } A) = \begin{bmatrix} 1 & -1 & 2 \\ 3 & 0 & -2 \\ 1 & 0 & 3 \end{bmatrix} \begin{bmatrix} 0 & 3 & 2 \\ -11 & 1 & 8 \\ 0 & -1 & 3 \end{bmatrix} = \begin{bmatrix} 11 & 0 & 0 \\ 0 & 11 & 0 \\ 0 & 0 & 11 \end{bmatrix}$$

$$= 11 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = 11 I = |A| I = \text{RHS}$$

Find inverse of following :

$$5] A = \begin{bmatrix} 2 & -2 \\ 4 & 3 \end{bmatrix} \quad A^{-1} = \frac{1}{|A|} \text{adj} A = \frac{1}{14} \begin{bmatrix} 3 & 2 \\ -4 & 2 \end{bmatrix}$$

$$6] A = \begin{bmatrix} -1 & 5 \\ -3 & 2 \end{bmatrix} \quad A^{-1} = \frac{1}{|A|} \text{adj} A = \frac{1}{13} \begin{bmatrix} 2 & -5 \\ 3 & -1 \end{bmatrix}$$

$$12] A = \begin{bmatrix} 3 & 7 \\ 2 & 5 \end{bmatrix} \text{ and } B = \begin{bmatrix} 6 & 8 \\ 7 & 9 \end{bmatrix}. \text{ Verify that } (AB)^{-1} = B^{-1}A^{-1}$$

$$\text{LHS: } AB = \begin{bmatrix} 3 & 7 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 6 & 8 \\ 7 & 9 \end{bmatrix} = \begin{bmatrix} 67 & 87 \\ 47 & 61 \end{bmatrix}$$

$$(AB)^{-1} = \frac{1}{|AB|} \text{adj} AB = \frac{1}{-2} \begin{bmatrix} 61 & -87 \\ -47 & 67 \end{bmatrix} \rightarrow (1)$$

$$A^{-1} = \frac{1}{|A|} \text{adj} A = \frac{1}{-2} \begin{bmatrix} 5 & -7 \\ -2 & 3 \end{bmatrix}$$

$$B^{-1} = \frac{1}{|B|} \text{adj} B = \frac{1}{-2} \begin{bmatrix} 9 & -8 \\ -7 & 6 \end{bmatrix}$$

$$B^{-1}A^{-1} = \frac{1}{-2} \begin{bmatrix} 9 & -8 \\ -7 & 6 \end{bmatrix} \frac{1}{-2} \begin{bmatrix} 5 & -7 \\ -2 & 3 \end{bmatrix} = \frac{1}{-2} \begin{bmatrix} 61 & -87 \\ -47 & 67 \end{bmatrix} \rightarrow (2)$$

$$\text{from (1) \& (2) } (AB)^{-1} = B^{-1}A^{-1}$$

$$13] \text{ If } A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix} \text{ Show that } A^2 - 5A + 7I = O. \text{ Hence find } A^{-1}.$$

$$A^2 = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 8 & 5 \\ -5 & 3 \end{bmatrix}$$

$$5A = \begin{bmatrix} 15 & 5 \\ -5 & 10 \end{bmatrix}$$

$$7I = \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix}$$

$$\therefore A^2 - SA + 7I$$

$$= \begin{bmatrix} 8 & 5 \\ -5 & 3 \end{bmatrix} - \begin{bmatrix} 15 & 5 \\ -5 & 10 \end{bmatrix} + \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = \text{RHS}$$

Hence,

$$A^2 - SA + 7I = 0 \quad \text{post multiply by } A^{-1}$$

$$A^2 A^{-1} - S A A^{-1} + 7 I A^{-1} = 0 A^{-1}$$

$$A(AA^{-1}) - S(AA^{-1}) + 7A^{-1} = 0$$

$$A - SI + 7A^{-1} = 0$$

$$7A^{-1} = SI - A$$

$$A^{-1} = \frac{1}{7}(SI - A)$$

$$= \frac{1}{7} \left(\begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} - \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix} \right)$$

$$= \frac{1}{7} \begin{bmatrix} 3 & -3 \\ -1 & 3 \end{bmatrix}$$

Eg 26] $A = \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix}$ Show that $A^2 - 4A + I = 0$, Find A^{-1}

$$A^2 = \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 7 & 12 \\ 4 & 7 \end{bmatrix}$$

$$4A = \begin{bmatrix} 8 & 12 \\ 4 & 8 \end{bmatrix}$$

$$\therefore A^2 - 4A + I$$

$$= \begin{bmatrix} 7 & 12 \\ 4 & 7 \end{bmatrix} - \begin{bmatrix} 8 & 12 \\ 4 & 8 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = \text{RHS}$$

Hence,

$$A^2 - 4A + I = 0 \quad \text{post multiply by } A^{-1}$$

$$A^2 A^{-1} - 4 A A^{-1} + I A^{-1} = 0 A^{-1}$$

$$A(AA^{-1}) - 4(AA^{-1}) + A^{-1} = 0$$

$$A - 4I + A^{-1} = 0$$

$$A^{-1} = 4I - A$$

$$= \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix} - \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 2 & -3 \\ -1 & 2 \end{bmatrix}$$

15] $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & -1 & 3 \end{bmatrix}$. Show that $A^3 - 6A^2 + 5A + 11I = O$.
Hence, find A^{-1} .

Sol] $A^2 = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & -1 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & -1 & 3 \end{bmatrix} = \begin{bmatrix} 4 & 2 & 1 \\ -3 & 8 & -14 \\ 7 & -3 & 14 \end{bmatrix}$

$$A^3 = \begin{bmatrix} 4 & 2 & 1 \\ -3 & 8 & -14 \\ 7 & -3 & 14 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & -1 & 3 \end{bmatrix} = \begin{bmatrix} 8 & 7 & 1 \\ -23 & 27 & -69 \\ 32 & -13 & 58 \end{bmatrix}$$

$$6A^2 = 6 \begin{bmatrix} 4 & 2 & 1 \\ -3 & 8 & -14 \\ 7 & -3 & 14 \end{bmatrix} = \begin{bmatrix} 24 & 12 & 6 \\ -18 & 48 & -84 \\ 42 & -18 & 84 \end{bmatrix}$$

$$5A = 5 \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & -1 & 3 \end{bmatrix} = \begin{bmatrix} 5 & 5 & 5 \\ 5 & 10 & -15 \\ 10 & -5 & 15 \end{bmatrix}$$

$$11I = 11 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 11 & 0 & 0 \\ 0 & 11 & 0 \\ 0 & 0 & 11 \end{bmatrix}$$

$$\therefore A^3 - 6A^2 + 5A + 11I = O$$

$$= \begin{bmatrix} 8 & 7 & 1 \\ -23 & 27 & -69 \\ 32 & -13 & 58 \end{bmatrix} - \begin{bmatrix} 24 & 12 & 6 \\ -18 & 48 & -84 \\ 42 & -18 & 84 \end{bmatrix} + \begin{bmatrix} 5 & 5 & 5 \\ 5 & 10 & -15 \\ 10 & -5 & 15 \end{bmatrix} + \begin{bmatrix} 11 & 0 & 0 \\ 0 & 11 & 0 \\ 0 & 0 & 11 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = O$$

Hence, $A^3 - 6A^2 + 5A + 11I = O$; post multiply by A^{-1}

$$A^3 A^{-1} - 6A^2 A^{-1} + 5A A^{-1} + 11I A^{-1} = O A^{-1}$$

$$A^2 - 6A + 5I + 11A^{-1} = O$$

$$11A^{-1} = 6A - 5I - A^2$$

$$A^{-1} = \frac{1}{11} [6A - 5I - A^2]$$

$$A^{-1} = \frac{1}{11} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \text{ACHIEVER}$$

$$A^{-1} = \frac{1}{11} \begin{bmatrix} 6 & 6 & 6 \\ 6 & 12 & -18 \\ 12 & -6 & 18 \end{bmatrix} - \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix} - \begin{bmatrix} 4 & 2 & 1 \\ -3 & 8 & -14 \\ 7 & -3 & 14 \end{bmatrix}$$

$$A^{-1} = \frac{1}{11} \begin{bmatrix} -3 & 4 & 5 \\ 9 & -1 & -4 \\ 5 & -3 & -1 \end{bmatrix}$$

16] $A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$. Verify that $A^3 - 6A^2 + 9A - 4I = 0$ and hence find A^{-1}

Sol] $A^2 = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} = \begin{bmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{bmatrix}$

$$A^3 = \begin{bmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{bmatrix} \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} = \begin{bmatrix} 22 & -21 & -21 \\ -21 & 22 & -21 \\ 21 & -21 & 22 \end{bmatrix}$$

$$6A^2 = 6 \begin{bmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{bmatrix} = \begin{bmatrix} 36 & -30 & 30 \\ -30 & 36 & -30 \\ 30 & -30 & 36 \end{bmatrix}$$

$$9A = 9 \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} = \begin{bmatrix} 18 & -9 & 9 \\ -9 & 18 & -9 \\ 9 & -9 & 18 \end{bmatrix}$$

$$4I = 4 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

$$\begin{aligned} \text{LHS} &= \therefore A^3 - 6A^2 + 9A - 4I \\ &= \begin{bmatrix} 22 & -21 & -21 \\ -21 & 22 & -21 \\ 21 & -21 & 22 \end{bmatrix} - \begin{bmatrix} 36 & -30 & 30 \\ -30 & 36 & -30 \\ 30 & -30 & 36 \end{bmatrix} + \begin{bmatrix} 18 & -9 & 9 \\ -9 & 18 & -9 \\ 9 & -9 & 18 \end{bmatrix} - \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = 0 = \text{RHS} \end{aligned}$$

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Hence, $A^3 - 6A^2 + 9A - 4I = 0$ post multiply by A^{-1}

$$A^3 A^{-1} - 6A^2 A^{-1} + 9A A^{-1} - 4I A^{-1} = 0 A^{-1}$$

$$A^2 - 6A + 9I - 4A^{-1} = 0$$

$$4A^{-1} = A^2 - 6A + 9I$$

$$A^{-1} = \frac{1}{4} [A^2 - 6A + 9I]$$

$$A^{-1} = \frac{1}{4} \left[\begin{pmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{pmatrix} - \begin{pmatrix} 12 & -6 & 6 \\ -6 & 12 & -6 \\ 6 & -6 & 12 \end{pmatrix} + \begin{pmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{pmatrix} \right]$$

$$A^{-1} = \frac{1}{4} \begin{pmatrix} 3 & 1 & -1 \\ 1 & 3 & 1 \\ -1 & 1 & 3 \end{pmatrix}$$

Find inverses of the following:

$$1] \quad A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 2 & 4 \\ 0 & 0 & 5 \end{pmatrix}$$

$$C_A = \begin{pmatrix} 24 & 0 & 4 \\ 0 & 5 & 0 \\ 23 & 1 & 2 \\ 0 & 5 & 0 \\ 23 & 1 & 2 \\ 24 & 0 & 4 \end{pmatrix} = \begin{pmatrix} 10 & 0 & 0 \\ -10 & 5 & 0 \\ 2 & -4 & 2 \end{pmatrix}$$

$$|A| = -10 + 0 + 0 = -10; \text{adj } A = C_A' = \begin{pmatrix} 10 & -10 & 2 \\ 0 & 5 & -4 \\ 0 & 0 & 2 \end{pmatrix}$$

$$\therefore A^{-1} = \frac{1}{|A|} (\text{adj } A) = \frac{1}{-10} \begin{pmatrix} 10 & -10 & 2 \\ 0 & 5 & -4 \\ 0 & 0 & 2 \end{pmatrix}$$

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$$8] A = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 3 & 0 \\ 5 & 2 & -1 \end{bmatrix}$$

$$C_A = \begin{bmatrix} 3 \cdot 0 & 3 \cdot 0 & 3 \cdot 3 \\ 2 \cdot -1 & 5 \cdot -1 & 5 \cdot 2 \\ 0 \cdot 0 & 1 \cdot 0 & 1 \cdot 0 \\ 2 \cdot -1 & 5 \cdot -1 & 5 \cdot 2 \\ 0 \cdot 0 & 1 \cdot 0 & 1 \cdot 0 \\ 3 \cdot 0 & 3 \cdot 0 & 3 \cdot 3 \end{bmatrix} = \begin{bmatrix} -3 & 3 & -9 \\ 0 & -1 & -2 \\ 0 & 0 & 3 \end{bmatrix}$$

$$|A| = -3 + 0 + 0 = -3; \text{adj } A = C_A' = \begin{bmatrix} -3 & 0 & 0 \\ 3 & -1 & 0 \\ -9 & -2 & 3 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} \text{adj } A = \frac{1}{-3} \begin{bmatrix} -3 & 0 & 0 \\ 3 & -1 & 0 \\ -9 & -2 & 3 \end{bmatrix}$$

$$9] A = \begin{bmatrix} 2 & 1 & 3 \\ 4 & -1 & 0 \\ -7 & 2 & 1 \end{bmatrix}$$

$$C_A = \begin{bmatrix} -1 \cdot 0 & 4 \cdot 0 & 4 \cdot -1 \\ 2 \cdot 1 & -7 \cdot 1 & -7 \cdot 2 \\ 1 \cdot 3 & 2 \cdot 3 & 2 \cdot 1 \\ 2 \cdot 1 & -7 \cdot 1 & -7 \cdot 2 \\ 1 \cdot 3 & 2 \cdot 3 & 2 \cdot 1 \\ -1 \cdot 0 & 4 \cdot 0 & 4 \cdot -1 \end{bmatrix} = \begin{bmatrix} -1 & -4 & 1 \\ 5 & 23 & -11 \\ 3 & 12 & -6 \end{bmatrix}$$

$$|A| = -2 - 4 + 3 = -5; \text{adj } A = C_A' = \begin{bmatrix} -1 & 5 & 3 \\ -4 & 23 & 12 \\ 1 & -11 & -6 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} (\text{adj } A) = \frac{1}{-5} \begin{bmatrix} -1 & 5 & 3 \\ -4 & 23 & 12 \\ 1 & -11 & -6 \end{bmatrix}$$

10] $A = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 2 & -3 \\ 3 & -2 & 4 \end{bmatrix}$

$$C_A = \begin{bmatrix} 2 & -3 & 0 & -3 & 0 & 2 \\ -1 & 2 & 1 & 2 & 1 & -1 \\ -2 & 4 & 3 & 4 & 3 & -2 \\ -1 & 2 & 1 & 2 & 1 & -1 \\ +2 & -3 & 0 & -3 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 2 & -9 & -6 \\ 0 & -2 & -1 \\ -1 & 3 & 2 \end{bmatrix}$$

$$|A| = 2 + 9 - 12 = -1; \text{adj} A = C_A' = \begin{bmatrix} 2 & 0 & -1 \\ -9 & -2 & 3 \\ -6 & -1 & 2 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} \text{adj} A = -1 \begin{bmatrix} 2 & 0 & -1 \\ -9 & -2 & 3 \\ -6 & -1 & 2 \end{bmatrix} = \begin{bmatrix} -2 & 0 & 1 \\ 9 & 2 & -3 \\ 6 & 1 & -2 \end{bmatrix}$$

11] $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos x & \sin x \\ 0 & \sin x & -\cos x \end{bmatrix}$

$$C_A = \begin{bmatrix} \cos x & \sin x & 0 & \sin x & 0 & \cos x \\ \sin x & -\cos x & 0 & -\cos x & 0 & \sin x \\ 0 & 0 & 1 & 0 & 1 & 0 \\ \sin x & -\cos x & 0 & -\cos x & 0 & \sin x \\ 0 & 0 & 1 & 0 & 1 & 0 \\ \cos x & \sin x & 0 & \sin x & 0 & \cos x \end{bmatrix} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -\cos x & -\sin x \\ 0 & -\sin x & \cos x \end{bmatrix}$$

$$|A| = -1 + 0 + 0 = -1; \text{adj} A = C_A' = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -\cos x & -\sin x \\ 0 & -\sin x & \cos x \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} (\text{adj} A) = -1 \begin{bmatrix} -1 & 0 & 0 \\ 0 & -\cos x & -\sin x \\ 0 & -\sin x & \cos x \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos x & \sin x \\ 0 & \sin x & -\cos x \end{bmatrix}$$

ACHIEVER

EXERCISE 4.6

Solve system of linear equations using matrix method.

7) $5x + 2y = 4$ $7x + 3y = 5$

$AX = B \rightarrow \textcircled{1}$

$$\begin{bmatrix} 5 & 2 \\ 7 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} \text{adj} A = \frac{1}{1} \begin{bmatrix} 3 & -2 \\ -7 & 5 \end{bmatrix}$$

$$\textcircled{1} \Rightarrow X = A^{-1}B$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3 & -2 \\ -7 & 5 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 12 - 10 \\ -28 + 25 \end{bmatrix} = \begin{bmatrix} 2 \\ -3 \end{bmatrix}$$

$$x = 2, y = -3$$

8) $2x - y = -2$ $3x + 4y = 3$

$AX = B \rightarrow \textcircled{1}$

$$\begin{bmatrix} 2 & -1 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -2 \\ 3 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} \text{adj} A = \frac{1}{11} \begin{bmatrix} 4 & 1 \\ -3 & 2 \end{bmatrix}$$

$$\textcircled{1} \Rightarrow X = A^{-1}B$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{11} \begin{bmatrix} 4 & 1 \\ -3 & 2 \end{bmatrix} \begin{bmatrix} -2 \\ 3 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{11} \begin{bmatrix} -5 \\ 12 \end{bmatrix} = \begin{bmatrix} -\frac{5}{11} \\ \frac{12}{11} \end{bmatrix}$$

$$x = -\frac{5}{11}, y = \frac{12}{11}$$

9] $4x - 3y = 3$, $3x - 5y = 7$ $\Rightarrow$ $\begin{bmatrix} 4 & -3 \\ 3 & -5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3 \\ 7 \end{bmatrix}$ $\Rightarrow$ $AX = B \rightarrow$ ①

$AX = B \rightarrow$ ①

$$\begin{bmatrix} 4 & -3 \\ 3 & -5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3 \\ 7 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} \text{adj} A = \frac{1}{-11} \begin{bmatrix} -5 & 3 \\ -3 & 4 \end{bmatrix}$$

① $\Rightarrow X = A^{-1}B$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{-11} \begin{bmatrix} -5 & 3 \\ -3 & 4 \end{bmatrix} \begin{bmatrix} 3 \\ 7 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \frac{-1}{11} \begin{bmatrix} 6 \\ 19 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -\frac{6}{11} \\ -\frac{19}{11} \end{bmatrix}$$

$$x = -\frac{6}{11}, y = -\frac{19}{11}$$

10] $5x + 2y = 3$, $3x + 2y = 5$

$AX = B \rightarrow$ ①

$$\begin{bmatrix} 5 & 2 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} (\text{adj} A) = \frac{1}{4} \begin{bmatrix} 2 & -2 \\ -3 & 5 \end{bmatrix}$$

① $\Rightarrow X = A^{-1}B$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 2 & -2 \\ -3 & 5 \end{bmatrix} \begin{bmatrix} 3 \\ 5 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 6 - 10 \\ -9 + 25 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -1 \\ 4 \end{bmatrix}$$

$$x = -1, y = 4$$

ACHIEVER

Eg 28] $3x - 2y + 3z = 8$

$$2x + y - z = 1$$

$$4x - 3y + 2z = 4$$

Sol] $A X = B \rightarrow \text{①}$

$$\begin{bmatrix} 3 & -2 & 3 \\ 2 & 1 & -1 \\ 4 & -3 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 8 \\ 1 \\ 4 \end{bmatrix}$$

$$C_A = \begin{bmatrix} 1 & -1 & 2 & -1 & 2 & 1 \\ -3 & 2 & 4 & 2 & 4 & -3 \\ -2 & 3 & 3 & 3 & 3 & -2 \\ -3 & 2 & 4 & 2 & 4 & -3 \\ -2 & 3 & 3 & 3 & 3 & -2 \\ 1 & -1 & 2 & -1 & 2 & 1 \end{bmatrix} = \begin{bmatrix} -1 & -8 & -10 \\ -5 & -6 & 1 \\ -1 & 9 & 7 \end{bmatrix}$$

$$|A| = -3 + 16 - 30 = -17 ; \text{adj } A = C_A^T = \begin{bmatrix} -1 & -5 & -1 \\ -8 & -6 & 9 \\ -10 & 1 & 7 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} \text{adj } A = \frac{1}{-17} \begin{bmatrix} -1 & -5 & -1 \\ -8 & -6 & 9 \\ -10 & 1 & 7 \end{bmatrix}$$

$$\text{①} \Rightarrow X = A^{-1}B = \frac{-1}{17} \begin{bmatrix} -1 & -5 & -1 \\ -8 & -6 & 9 \\ -10 & 1 & 7 \end{bmatrix} \begin{bmatrix} 8 \\ 1 \\ 4 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{-1}{17} \begin{bmatrix} -8-5-4 \\ -64-6+36 \\ -80+1+28 \end{bmatrix} = \frac{-1}{17} \begin{bmatrix} -17 \\ -34 \\ -51 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$x=1, y=2, z=3$$

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Eg 29] Let the no.s be x, y, z

$$\begin{aligned} x + y + z &= 6 \\ y + 3z &= 11 \\ x + z &= 2y \quad \text{or} \quad x - 2y + z = 0 \end{aligned}$$

$$A \cdot X = B \rightarrow \textcircled{1}$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 3 \\ 1 & -2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 11 \\ 0 \end{bmatrix}$$

$$C_A = \begin{bmatrix} 1 & 3 & 0 & 3 & 0 & 1 \\ -2 & 1 & 1 & 1 & 1 & -2 \\ -2 & 1 & 1 & 1 & 1 & -2 \\ 1 & 3 & 0 & 3 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 7 & 3 & -1 \\ -3 & 0 & 3 \\ 2 & -3 & 1 \end{bmatrix}$$

$$|A| = 7 + 3 - 1 = 9; \quad \text{adj } A = C_A' = \begin{bmatrix} 7 & -3 & 2 \\ 3 & 0 & -3 \\ -1 & 3 & 1 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} \text{adj } A = \frac{1}{9} \begin{bmatrix} 7 & -3 & 2 \\ 3 & 0 & -3 \\ -1 & 3 & 1 \end{bmatrix}$$

$$\textcircled{1} \Rightarrow X = A^{-1}B$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{9} \begin{bmatrix} 7 & -3 & 2 \\ 3 & 0 & -3 \\ -1 & 3 & 1 \end{bmatrix} \begin{bmatrix} 6 \\ 11 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{9} \begin{bmatrix} 42 - 33 + 0 \\ 18 + 0 + 0 \\ -6 + 33 + 0 \end{bmatrix} = \frac{1}{9} \begin{bmatrix} 9 \\ 18 \\ 27 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$x = 1, y = 2, z = 3$$

ACHIEVER

- 16] The cost of 4 kg onion, 3 kg wheat and 2 kg rice is Rs 60.
 The cost of 2 kg onion, 4 kg wheat and 6 kg rice is Rs 90.
 The cost of 6 kg onion, 2 kg wheat and 3 kg rice is Rs 70.
 Find cost of each item per kg by matrix method.

Sol] Let onion, wheat, rice be x, y, z respectively.

$$\begin{aligned} \text{According to data, } 4x + 3y + 2z &= 60 \\ 2x + 4y + 6z &= 90 \\ 6x + 2y + 3z &= 70 \end{aligned}$$

$$AX = B \rightarrow \text{①}$$

$$\begin{bmatrix} 4 & 3 & 2 \\ 2 & 4 & 6 \\ 6 & 2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 60 \\ 90 \\ 70 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} \begin{bmatrix} 0 & 30 & -20 \\ -5 & 0 & 10 \\ 10 & -20 & 10 \end{bmatrix}$$

$$|A| = \begin{vmatrix} 4 & 3 & 2 \\ 2 & 4 & 6 \\ 6 & 2 & 3 \end{vmatrix} = 0 + 90 - 40 = 50$$

$$A^{-1} = \frac{1}{|A|} (\text{adj } A) = \frac{1}{50} \begin{bmatrix} 0 & -5 & 10 \\ 30 & 0 & -20 \\ -20 & 10 & 10 \end{bmatrix}$$

$$\text{①} \Rightarrow X = A^{-1}B$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{50} \begin{bmatrix} 0 & -5 & 10 \\ 30 & 0 & -20 \\ -20 & 10 & 10 \end{bmatrix} \begin{bmatrix} 60 \\ 90 \\ 70 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{50} \begin{bmatrix} 0 - 450 + 700 \\ 1800 - 0 - 1400 \\ -1200 + 900 + 700 \end{bmatrix} = \frac{1}{50} \begin{bmatrix} 250 \\ 400 \\ 400 \end{bmatrix} = \begin{bmatrix} 5 \\ 8 \\ 8 \end{bmatrix}$$

Cost of 1 kg of onion = ₹ 5

Cost of 1 kg of wheat = ₹ 8

cost of 1 kg of rice = ₹ 8

15] $A = \begin{bmatrix} 2 & -3 & 5 \\ 3 & 2 & -4 \\ 1 & 1 & -2 \end{bmatrix}$, find A^{-1} . Using A^{-1} solve the system of equations: $2x - 3y + 5z = 11$, $3x + 2y - 4z = -5$, $x + y - 2z = -3$

Sol] $C_A = \begin{bmatrix} 2 & -4 & 3 & -4 & 3 & 2 \\ 1 & -2 & 1 & -2 & 1 & 1 \\ -3 & 5 & 2 & 5 & 2 & -3 \\ 1 & -2 & 1 & -2 & 1 & 1 \\ -3 & 5 & 2 & 5 & 2 & -3 \\ 2 & -4 & 3 & -4 & 3 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 2 & 1 \\ -1 & -9 & -5 \\ 2 & 23 & 13 \end{bmatrix}$

$|A| = 0 - 6 + 5 = -1$; $\text{adj } A = C_A' = \begin{bmatrix} 0 & -1 & 2 \\ 2 & -9 & 23 \\ 1 & -5 & 13 \end{bmatrix}$

$A^{-1} = \frac{1}{|A|} (\text{adj } A) = -1 \begin{bmatrix} 0 & -1 & 2 \\ 2 & -9 & 23 \\ 1 & -5 & 13 \end{bmatrix} = \begin{bmatrix} 0 & 1 & -2 \\ -2 & 9 & -23 \\ -1 & 5 & -13 \end{bmatrix}$

Given equations: $2x - 3y + 5z = 11$
 $3x + 2y - 4z = -5$
 $x + y - 2z = -3$

$A X = B \rightarrow \text{①}$

$\begin{bmatrix} 2 & -3 & 5 \\ 3 & 2 & -4 \\ 1 & 1 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 11 \\ -5 \\ -3 \end{bmatrix}$

① $\Rightarrow X = A^{-1} B$

$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 & 1 & -2 \\ -2 & 9 & -23 \\ -1 & 5 & -13 \end{bmatrix} \begin{bmatrix} 11 \\ -5 \\ -3 \end{bmatrix}$

$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 - 5 + 6 \\ -22 - 45 + 69 \\ -11 - 25 + 39 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$

$x = 1, y = 2, z = 3$

ACHIEVER

$$11) \quad 2x + y + z = 1$$

$$x - 2y - z = \frac{3}{2}$$

$$3y - 5z = 9$$

$$\text{sol]} \quad A X = B \rightarrow \textcircled{1}$$

$$\begin{bmatrix} 2 & 1 & 1 \\ 1 & -2 & -1 \\ 0 & 3 & -5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ \frac{3}{2} \\ 9 \end{bmatrix}$$

$$C_A = \begin{bmatrix} -2 & -1 & 1 \\ 3 & -5 & 0 \\ 1 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 13 & 5 & 3 \\ 8 & -10 & -6 \\ 1 & 3 & -5 \end{bmatrix}$$

$$|A| = 26 + 5 + 3 = 34; \quad \text{adj } A = C_A' = \begin{bmatrix} 13 & 8 & 1 \\ 5 & -10 & 3 \\ 3 & -6 & -5 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} \text{adj } A = \frac{1}{34} \begin{bmatrix} 13 & 8 & 1 \\ 5 & -10 & 3 \\ 3 & -6 & -5 \end{bmatrix}$$

$$\textcircled{1} \Rightarrow X = A^{-1} B$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{34} \begin{bmatrix} 13 & 8 & 1 \\ 5 & -10 & 3 \\ 3 & -6 & -5 \end{bmatrix} \begin{bmatrix} 1 \\ \frac{3}{2} \\ 9 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{34} \begin{bmatrix} 34 \\ 17 \\ -51 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ \frac{1}{2} \\ -\frac{3}{2} \end{bmatrix}$$

$$x = 1, y = \frac{1}{2}, z = -\frac{3}{2}$$

$$12] \quad x - y + z = 4$$

$$2x + y - 3z = 0$$

$$x + y + z = 2$$

$$\text{Sol]} \quad A X = B \rightarrow \textcircled{1}$$

$$\begin{bmatrix} 1 & -1 & 1 \\ 2 & 1 & -3 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 0 \\ 2 \end{bmatrix}$$

$$C_A = \begin{bmatrix} 1 & -3 & 2 & -3 & 2 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ -1 & 1 & 1 & 1 & -1 & -1 \\ 1 & 1 & 1 & 1 & 1 & -1 \\ -1 & 1 & 1 & 1 & 1 & -1 \\ 1 & -3 & 2 & -3 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 4 & -5 & 1 \\ 2 & 0 & -2 \\ 2 & 5 & 3 \end{bmatrix}$$

$$|A| = 4 + 5 + 1 = 10 ; \text{adj} A = C_A' = \begin{bmatrix} 4 & 2 & 2 \\ -5 & 0 & 5 \\ 1 & -2 & 3 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} (\text{adj} A) = \frac{1}{10} \begin{bmatrix} 4 & 2 & 2 \\ -5 & 0 & 5 \\ 1 & -2 & 3 \end{bmatrix}$$

$$\textcircled{1} \Rightarrow X = A^{-1} B$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{10} \begin{bmatrix} 4 & 2 & 2 \\ -5 & 0 & 5 \\ 1 & -2 & 3 \end{bmatrix} \begin{bmatrix} 4 \\ 0 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{10} \begin{bmatrix} 16 + 0 + 4 \\ -20 + 0 + 10 \\ -4 + 0 + 6 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$$

$$x = 2, y = -1, z = 1$$

Achiever

$$\begin{aligned} 13] \quad & 2x + 3y + 3z = 5 \\ & x - 2y + z = -4 \\ & 3x - y - 2z = 3 \end{aligned}$$

$$\text{sd]} \quad AX = B \rightarrow \textcircled{1}$$

$$\begin{bmatrix} 2 & 3 & 3 \\ 1 & -2 & 1 \\ 3 & -1 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 5 \\ -4 \\ 3 \end{bmatrix}$$

$$C_A = \begin{bmatrix} -2 & 1 & 1 \\ -1 & -2 & 3 \\ 3 & 3 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 & -2 \\ 3 & -1 & 3 \\ -1 & -2 & 3 \end{bmatrix} = \begin{bmatrix} 5 & 5 & 5 \\ 3 & -13 & 11 \\ 9 & 1 & -7 \end{bmatrix}$$

$$|A| = 10 + 15 + 15 = 40; \quad \text{adj } A = C_A' = \begin{bmatrix} 5 & 3 & 9 \\ 5 & -13 & 1 \\ 5 & 11 & -7 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} \text{adj } A = \frac{1}{40} \begin{bmatrix} 5 & 3 & 9 \\ 5 & -13 & 1 \\ 5 & 11 & -7 \end{bmatrix}$$

$$\textcircled{1} \Rightarrow X = A^{-1}B$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{40} \begin{bmatrix} 5 & 3 & 9 \\ 5 & -13 & 1 \\ 5 & 11 & -7 \end{bmatrix} \begin{bmatrix} 5 \\ -4 \\ 3 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{40} \begin{bmatrix} 25 - 12 + 27 \\ 25 + 52 + 3 \\ 25 - 44 - 21 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$$

$$x = 1, y = 2, z = -1$$

$$14] \quad x - y + 2z = 7$$

$$3x + 4y - 5z = -5$$

$$2x - y + 3z = 12$$

$$\text{Sol.}] \quad AX = B \rightarrow (1)$$

$$\begin{pmatrix} 1 & -1 & 2 \\ 3 & 4 & -5 \\ 2 & -1 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 7 \\ -5 \\ 12 \end{pmatrix}$$

$$C_A = \begin{vmatrix} 4 & -5 & 3 & -5 & 3 & 4 \\ -1 & 3 & 2 & 3 & 2 & -1 \\ -1 & 2 & 1 & 2 & 1 & -1 \\ -1 & 3 & 2 & 3 & 2 & -1 \\ -1 & 2 & 1 & 2 & 1 & -1 \\ 4 & -5 & 3 & -5 & 3 & 4 \end{vmatrix} = \begin{pmatrix} 7 & -19 & -11 \\ -1 & -1 & -1 \\ -3 & -11 & 7 \end{pmatrix}$$

$$|A| = 7 + 19 - 22 = 4; \quad \text{adj} A = C_A' = \begin{pmatrix} 7 & 1 & -3 \\ -19 & -1 & 11 \\ -11 & -1 & 7 \end{pmatrix}$$

$$A^{-1} = \frac{1}{|A|} (\text{adj} A) = \frac{1}{4} \begin{pmatrix} 7 & 1 & -3 \\ -19 & -1 & 11 \\ -11 & -1 & 7 \end{pmatrix}$$

$$(1) \Rightarrow X = A^{-1}B$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \frac{1}{4} \begin{pmatrix} 7 & 1 & -3 \\ -19 & -1 & 11 \\ -11 & -1 & 7 \end{pmatrix} \begin{pmatrix} 7 \\ -5 \\ 12 \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \frac{1}{4} \begin{pmatrix} 49 - 5 - 36 \\ -133 + 5 + 132 \\ -77 + 5 + 84 \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}$$

$$x = 2, y = 1, z = 3$$

13] Solve by matrix method:

$$2x + 3y + 3z = 5$$

$$x - 2y + z = -4$$

$$3x - y - 2z = 3$$

$$AX = B \rightarrow \textcircled{1}$$

$$\begin{bmatrix} 2 & 3 & 3 \\ 1 & -2 & 1 \\ 3 & -1 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 5 \\ -4 \\ 3 \end{bmatrix}$$

$$C_A = \begin{bmatrix} -2 & 1 & 1 & 1 & -2 \\ -1 & -2 & 3 & -2 & 3 & -1 \\ 3 & 3 & 2 & 3 & 2 & 3 \\ -1 & -2 & 3 & -2 & 3 & -1 \\ 3 & 3 & 2 & 3 & 2 & 3 \\ -2 & 1 & 1 & 1 & -2 \end{bmatrix} = \begin{bmatrix} 5 & 5 & 5 \\ 3 & -13 & 11 \\ 9 & 1 & -7 \end{bmatrix}$$

$$|A| = 10 + 15 + 15 = 40 ; \text{adj} A = C_A' = \begin{bmatrix} 5 & 3 & 9 \\ 5 & -13 & 1 \\ 5 & 11 & -7 \end{bmatrix}$$

$$A^{-1} = \frac{1}{40} \begin{bmatrix} 5 & 3 & 9 \\ 5 & -13 & 1 \\ 5 & 11 & -7 \end{bmatrix}$$

$$X = A^{-1}B$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{40} \begin{bmatrix} 5 & 3 & 9 \\ 5 & -13 & 1 \\ 5 & 11 & -7 \end{bmatrix} \begin{bmatrix} 5 \\ -4 \\ 3 \end{bmatrix}$$

$$= \frac{1}{40} \begin{bmatrix} 25 - 12 + 27 \\ 25 + 52 + 3 \\ 25 - 44 - 21 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$$

$$x = 1, y = 2, z = -1$$

EXERCISE 4.1

Evaluate the determinants:

$$1) \begin{vmatrix} 2 & 4 \\ -5 & -1 \end{vmatrix} = -2 + 20 = 18$$

$$2) \text{ i) } \begin{vmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{vmatrix} = \cos^2 \theta + \sin^2 \theta = 1$$

$$\text{ii) } \begin{vmatrix} x^2 - x + 1 & x - 1 \\ x + 1 & x + 1 \end{vmatrix} = (x+1)(x^2 - x + 1) - (x^2 - 1)$$

$$= x^3 + 1 - x^2 + 1 - x^2 + 1 = x^3 - x^2 + 2$$

$$3) \text{ If } A = \begin{bmatrix} 1 & 2 \\ 4 & 2 \end{bmatrix} \text{ show that } |2A| = 4|A|$$

$$|2A| = 2^2 |A| = 4(2-8) = -24$$

$$4) \text{ If } A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 4 \end{bmatrix}, \text{ show that } |3A| = 27|A|$$

$$|3A| = 3^3 |A| = 27[1(4-0) + 0 + 1(0-0)]$$

$$= 27(4) = 108$$

5) Evaluate the determinants:

$$\text{i) } \begin{vmatrix} 3 & -1 & -2 \\ 0 & 0 & -1 \\ 3 & -5 & 0 \end{vmatrix} = 3(0-5) + 1(3) - 2(0)$$

$$= -15 + 3 = -12$$

$$\text{ii) } \begin{vmatrix} 3 & -4 & 5 \\ 1 & 1 & -2 \\ 2 & 3 & 1 \end{vmatrix} = 3(1+6) + 4(1+4) + 5(3-2)$$

$$= 21 + 20 + 5 = 46$$

Achiever

$$\text{iii)] } \begin{vmatrix} 0 & 1 & 2 \\ -1 & 0 & -3 \\ -2 & 3 & 0 \end{vmatrix} = 0(0+9) - 1(0-6) + 2(-3+0) \\ = 0 + 6 - 6 \\ = 0$$

$$\text{iv)] } \begin{vmatrix} 2 & -1 & -2 \\ 0 & 2 & -1 \\ 3 & -5 & 0 \end{vmatrix} = 2(0-5) + 1(0+3) - 2(0-6) \\ = -10 + 3 + 12 \\ = 5$$

$$6) \text{ If } A = \begin{bmatrix} 1 & 1 & -2 \\ 2 & 1 & -3 \\ 5 & 4 & -9 \end{bmatrix}, \text{ find } |A| \\ = 1(-9+12) - 1(-18+15) - 2(8-5) \\ = 3 + 3 - 6$$

7) Find values of x if:

$$\text{i)] } \begin{vmatrix} 2 & 4 \\ 5 & 1 \end{vmatrix} = \begin{vmatrix} 2x & 4 \\ 6 & x \end{vmatrix}$$

$$2-20 = 2x^2 - 24$$

$$-18+24 = 2x^2$$

$$6 = 2x^2 = |A|^2 = |A|^2$$

$$x = \pm\sqrt{3}$$

$$\text{ii)] } \begin{vmatrix} 2 & 3 \\ 4 & 5 \end{vmatrix} = \begin{vmatrix} x & 3 \\ 2x & 5 \end{vmatrix}$$

$$10-12 = 5x-6x$$

$$-2 = -x$$

$$x = 2$$

$$8) \text{ If } \begin{vmatrix} x & 2 \\ 18 & x \end{vmatrix} = \begin{vmatrix} 6 & 2 \\ 18 & 6 \end{vmatrix} \quad x = ?$$

$$(A) 6 \quad (B) \pm 6 \quad (C) 6 \quad (D) 0$$

$$x^2 - 36 = 36 - 36$$

$$x^2 = 36$$

$$x = \pm 6$$

Area of a triangle:

*] If 3 points are collinear
area of $\Delta = 0$

$$*] \text{ Area of } \Delta = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

*] Area of $\Delta = \pm K$ if unknown value exists in vertices

*] Equation of a line if 2 pts are given $A(x_1, y_1)$, $B(x_2, y_2)$

Step 1: $P(x, y)$ be any point on the line AB.

Step 2: Now pts A, B, P are collinear

$\therefore$ area of $\Delta = 0$

EXERCISE 4.3

1] Find the area of Δ with vertices given,

(i) $(1, 0)$, $(6, 0)$, $(4, 3)$
 x_1, y_1 x_2, y_2 x_3, y_3

$$\begin{aligned} \text{An of } \Delta ABC &= \frac{1}{2} \begin{vmatrix} 1 & 0 & 1 \\ 6 & 0 & 1 \\ 4 & 3 & 1 \end{vmatrix} = \frac{1}{2} [1(0-3) - 0 + 1(18-0)] \\ &= \frac{1}{2} (-3+18) = \frac{15}{2} \text{ square unit} \end{aligned}$$

ii) $(2, 7)$, $(1, 1)$, $(10, 8)$
 A B C

$$\begin{aligned} \text{An of } \Delta ABC &= \frac{1}{2} \begin{vmatrix} 2 & 7 & 1 \\ 1 & 1 & 1 \\ 10 & 8 & 1 \end{vmatrix} = \frac{1}{2} [2(1-8) - 7(8-10) + 1(8-10)] \\ &= \frac{1}{2} [-14 + 63 - 2] = \frac{47}{2} \text{ sq. unit} \end{aligned}$$

iii) $(-2, -3)$, $(3, 2)$, $(-1, -8)$
 A B C

$$\begin{aligned} \text{An of } \Delta ABC &= \frac{1}{2} \begin{vmatrix} -2 & -3 & 1 \\ 3 & 2 & 1 \\ -1 & -8 & 1 \end{vmatrix} = \frac{1}{2} [-2(2+8) + 3(3+1) + 1(-24+2)] \\ &= \frac{1}{2} (-20+12-22) = \frac{-42+12}{2} \\ &= \frac{-30}{2} = -15 = 15 \text{ sq. units} \end{aligned}$$

2] Show that the pts $A(a, b+c), B(b, c+a), C(c, a+b)$ are collinear

Sol] Consider area of $\Delta ABC = \frac{1}{2} \begin{vmatrix} a & b+c & 1 \\ b & c+a & 1 \\ c & a+b & 1 \end{vmatrix}$

$$C_2 \rightarrow C_2 + C_1$$

$$= \frac{1}{2} \begin{vmatrix} a & a+b+c & 1 \\ b & a+b+c & 1 \\ c & a+b+c & 1 \end{vmatrix}$$

$$= \frac{1}{2} (a+b+c) \begin{vmatrix} a & 1 & 1 \\ b & 1 & 1 \\ c & 1 & 1 \end{vmatrix}$$

$$\Delta = 0 (\because C_2 \sim C_3)$$

$\Rightarrow A, B, C$ are collinear.

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Page _____Consistency of system of equation :

If A is a non singular matrix, system of equations is consistent and has a unique solution.
ie, $|A| \neq 0$, unique solution and consistent.

If $|A| = 0$,

(a) $(\text{adj } A)B \neq 0$ then solution does not exist, system is inconsistent.

(b) $(\text{adj } A)B = 0$ then system may be consistent with infinite solutions or may be inconsistent with no solution.

EXERCISE 4.6

$$x + 2y = 2 ; 2x + 3y = 3$$

$$\begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

$$|A| = 3 - 4 = -1 \neq 0$$

$\Rightarrow$ system is consistent and has unique solution.

$$2x - y = 5 ; x + y = 4$$

$$\begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 5 \\ 4 \end{bmatrix}$$

$$|A| = 2 + 1 = 3 \neq 0$$

$\Rightarrow$ system is consistent and has unique solution.

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$$3) \quad x+3y=5; 2x+6y=8$$

$$\begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 5 \\ 8 \end{bmatrix}$$

$$|A| = 6 - 6 = 0$$

$$(\text{adj } A)B = \begin{bmatrix} 6 & -3 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 5 \\ 8 \end{bmatrix} = \begin{bmatrix} 6 \\ -2 \end{bmatrix} \neq 0$$

$\Rightarrow$ system is inconsistent and has no solution.

$$4) \quad x+y+z=1; 2x+3y+2z=2; ax+ay+2az=4$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 2 \\ a & a & 2a \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}$$

$$|A| = 1(6a-2a) - 1(4a-2a) + 1(2a-3a) \\ = 4a - 2a - a = a \neq 0$$

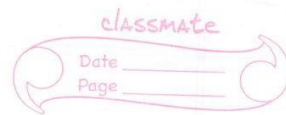
$\Rightarrow$ system is consistent and has unique solution.

$$5) \quad 3x-y-2z=2; 2y-z=-1; 3x-5y=3$$

$$\begin{bmatrix} 3 & -1 & -2 \\ 0 & 2 & -1 \\ 3 & -5 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$$

$$|A| = 3(0-5) + 1(0+3) - 2(0-6) \\ = -15 + 3 + 12 = 0$$

$$C_A = \begin{bmatrix} 2 & -1 & 0 & -1 & 0 & 2 \\ -5 & 0 & 3 & 0 & 3 & -5 \\ -1 & -2 & 3 & -2 & 3 & -1 \\ -5 & 0 & 3 & 0 & 3 & -5 \\ -1 & -2 & 3 & -2 & 3 & -1 \\ 2 & -1 & 0 & -1 & 0 & 2 \end{bmatrix} \rightarrow P = \begin{bmatrix} -5 & + & 1 \\ 10 & - & 1 \\ 5 & & \end{bmatrix}$$



$$(\text{adj } A) = C_A^T = \begin{bmatrix} -5 & 10 & 5 \\ \end{bmatrix}$$

$$\therefore (\text{adj } A) B = \begin{bmatrix} -5 & 10 & 5 \\ \end{bmatrix}_{3 \times 3} \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}_{3 \times 1} = \begin{bmatrix} -5 \\ \end{bmatrix} \neq 0$$

$\Rightarrow$ system is inconsistent and has no solution.

6) $5x - y + 4z = 5$; $2x + 3y + 5z = 2$; $5x - 2y + 6z = -1$

$$\begin{bmatrix} 5 & -1 & 4 \\ 2 & 3 & 5 \\ 5 & -2 & 6 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \\ -1 \end{bmatrix}$$

$$\begin{aligned} |A| &= 5(18 + 10) + 1(12 - 25) + 4(-4 - 15) \\ &= 140 - 13 - 76 \\ &= 51 \neq 0 \end{aligned}$$

$\Rightarrow$ system is consistent and has unique solution.

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DETERMINANTSEXERCISE 4.33] Find the value of k if the area of triangle is 4 sq. units.

i] $(k, 0), (4, 0), (0, 2)$

Given: $\Delta ABC = 4$

$$\frac{1}{2} \begin{vmatrix} k & 0 & 1 \\ 4 & 0 & 1 \\ 0 & 2 & 1 \end{vmatrix} = 4$$

$$k(-2) + 1(8) = \pm 8$$

$$-2k + 8 = \pm 8$$

$$8 \mp 8 = 2k$$

$$2k = 0 \quad | \quad 2k = 8 + 8 = 16$$

$$k = 0 \quad | \quad k = 8$$

ii] $(-2, 0), (0, 4), (0, k)$

Given: $\Delta ABC = 4$

$$\frac{1}{2} \begin{vmatrix} -2 & 0 & 1 \\ 0 & 4 & 1 \\ 0 & k & 1 \end{vmatrix} = 4$$

$$-2(4-k) - 0 + 1(0-0) = \pm 8$$

$$-8 + 2k = \pm 8$$

$$2k = \pm 8 + 8$$

$$2k = 8 + 8 \quad | \quad 2k = -8 + 8$$

$$k = 8 \quad | \quad k = 0$$

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- 5] If area of Δ^k is 35 sq. units, with vertices $(2, -6)$, $(5, 4)$, $(k, 4)$ then $k =$

(A) 12 (B) -2 (C) -12, -2 (D) 12, -2

$$\Delta_{ABC} = \frac{1}{2} \begin{vmatrix} 2 & -6 & 1 \\ 5 & 4 & 1 \\ k & 4 & 1 \end{vmatrix} = 35$$

$$2(4 - 4) + 6(5 - k) + 1(20 - 4k) = \pm 70$$

$$0 + 30 - 6k + 20 - 4k = \pm 70$$

$$50 - 10k = \pm 70$$

$$10k = 50 \pm 70$$

$$10k = 50 + 70 \quad | \quad 10k = 50 - 70$$

$$k = 12 \quad | \quad k = -2$$

- 4] i] Find eq. of the line joining $(1, 2)$, $(3, 6)$ using determinants.

Let $A = (1, 2)$, $B = (3, 6)$, $P = (x, y)$ on $\overline{AB}$

$\therefore$ APB are collinear, $\Delta_{PAB} = 0$

$$\Rightarrow \frac{1}{2} \begin{vmatrix} x & y & 1 \\ 1 & 2 & 1 \\ 3 & 6 & 1 \end{vmatrix} = 0$$

$$x(2 - 6) - y(1 - 3) + 1(6 - 6) = 0$$

$$2x - 6x - y + 3y = 0$$

$$-4x + 2y = 0$$

$$4x - 2y = 0$$

$$\boxed{2x - y = 0}$$

- ii] Find the eq. of line joining $(3, 1)$, $(9, 3)$ using determinants

Let $A(3, 1)$, $B(9, 3)$, $P(x, y)$ be any pt. on $\overline{AB}$

$\therefore$ APB are collinear, $\Delta_{APB} = 0$

$$\Rightarrow \frac{1}{2} \begin{vmatrix} x & y & 1 \\ 3 & 1 & 1 \\ 9 & 3 & 1 \end{vmatrix} = 0$$

$$x(1-3) - y(3-9) + 1(9-9) = 0$$

$$x - 3x - 3y + 9y = 0$$

$$-2x + 6y = 0$$

$$-2x + 6y = 0$$

$$x - 3y = 0$$

EXERCISE 4.4

Write minors and co-factors of the elements of following determinants:

i] $\begin{vmatrix} 2 & -4 \\ 0 & 3 \end{vmatrix}$

$$M_{11} = \text{minor of } 2 = 3 ; A_{11} = (-1)^{1+1} 3 = 3$$

$$M_{12} = 0 ; A_{12} = -0 = 0$$

$$M_{21} = -4 ; A_{21} = -(-4) = 4$$

$$M_{22} = 2 ; A_{22} = +2 = 2$$

ii] $\begin{vmatrix} a & c \\ b & d \end{vmatrix}$

$$M_{11} = d ; A_{11} = d$$

$$M_{12} = b ; A_{12} = -b$$

$$M_{21} = c ; A_{21} = -c$$

$$M_{22} = a ; A_{22} = a$$

2] i] $\begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$

$$M_{11} = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1 ; A_{11} = +1 = (-1)^{1+1} M_{11}$$

$$M_{12} = 0 ; A_{12} = -0 = 0$$

$$M_{13} = 0 ; A_{13} = 0$$

$$M_{21} = 0 ; A_{21} = 0$$

$$M_{22} = 1 ; A_{22} = +1$$

$$M_{23} = 0 ; A_{23} = 0$$

$$M_{31} = 0 ; A_{31} = 0$$

$$M_{32} = 0 ; A_{32} = 0$$

$$M_{33} = 1 ; A_{33} = 1$$

$$C_A = \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

$$0 =$$

$$\text{ii) } A = \begin{bmatrix} 1 & 0 & 4 \\ 3 & 5 & -1 \\ 0 & 1 & 2 \end{bmatrix}$$

$$M_{11} = 11; A_{11} = 11$$

$$M_{31} = -20; A_{31} = -20$$

$$M_{12} = 6; A_{12} = -6$$

$$M_{32} = -13; A_{32} = +13$$

$$M_{13} = 3; A_{13} = 3$$

$$M_{33} = 5; A_{33} = 5$$

$$M_{21} = -4; A_{21} = +4$$

$$M_{22} = 2; A_{22} = 2$$

$$M_{23} = 1; A_{23} = -1$$

Properties under expansion of determinants:

Property i) Elements of any row or column multiplied by their own cofactors and added gives the value of the determinant.

3) Eg: Using cofactors of elements of 2nd row, evaluate $\Delta =$

$$\Delta = \begin{vmatrix} 5 & 3 & 8 \\ 2 & 0 & 1 \\ 1 & 2 & 3 \end{vmatrix}$$

extra along R_1 : $5 \begin{vmatrix} 0 & 1 \\ 2 & 3 \end{vmatrix} - 3 \begin{vmatrix} 2 & 1 \\ 1 & 3 \end{vmatrix} + 8 \begin{vmatrix} 2 & 0 \\ 1 & 2 \end{vmatrix}$

$$\Delta = 5(-2) - 3(5) + 8(4) = -10 - 15 + 32 = \boxed{7}$$

along R_2 : $\Delta = -2 \begin{vmatrix} 3 & 8 \\ 2 & 3 \end{vmatrix} + 0 - 1 \begin{vmatrix} 5 & 3 \\ 1 & 2 \end{vmatrix}$

$$\Delta = -2(-7) - 1(7) = 14 - 7 = \boxed{7}$$

extra along C_3 : $8 \begin{vmatrix} 2 & 0 \\ 1 & 2 \end{vmatrix} - 1 \begin{vmatrix} 5 & 3 \\ 1 & 2 \end{vmatrix} + 3 \begin{vmatrix} 5 & 3 \\ 2 & 0 \end{vmatrix}$

$$\Delta = 32 - 7 - 18 = \boxed{7}$$

Property ii) If elements of any row or column are multiplied by cofactors of another row or column, and added, their sum is always equal to zero.

Eg: Along R_1 and R_3 : $5 \begin{vmatrix} 3 & 8 \\ 0 & 1 \end{vmatrix} - 3 \begin{vmatrix} 5 & 8 \\ 2 & 1 \end{vmatrix} + 8 \begin{vmatrix} 5 & 3 \\ 2 & 0 \end{vmatrix}$

$$= 15 + 33 - 48$$

$$= 0$$

Eg: Along C_1 and C_2 :
$$= -5 \begin{vmatrix} 2 & 1 \\ 1 & 3 \end{vmatrix} + 2 \begin{vmatrix} 5 & 8 \\ 1 & 3 \end{vmatrix} - 1 \begin{vmatrix} 5 & 8 \\ 2 & 1 \end{vmatrix}$$

$$= -25 + 14 + 11 = -18 + 11 = -7$$

$$= 0$$

Theorem: If A is any square matrix of order n , then

$$A(\text{adj } A) = (\text{adj } A)A = |A|I \quad [\text{Adj } A = C_A']$$

Proof: Let $A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$; $C_A = \begin{pmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{pmatrix}$

$$\text{Adj } A = C_A' = \begin{pmatrix} A_{11} & A_{21} & A_{31} \\ A_{12} & A_{22} & A_{32} \\ A_{13} & A_{23} & A_{33} \end{pmatrix}$$

$$\text{LHS} = A \cdot \text{adj } A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \begin{pmatrix} A_{11} & A_{21} & A_{31} \\ A_{12} & A_{22} & A_{32} \\ A_{13} & A_{23} & A_{33} \end{pmatrix}$$

$$= \begin{pmatrix} |A| & 0 & 0 \\ 0 & |A| & 0 \\ 0 & 0 & |A| \end{pmatrix} = |A| \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = |A|I = \text{RHS}$$

Theorem: If A and B are non-singular matrices of same order, then AB and BA are also non singular matrices of same order.

Theorem: Determinant of product of matrices is equal to the product of their respective determinant.

$$\boxed{|AB| = |A||B|}$$

Eg: If $A = \begin{bmatrix} 2 & -1 \\ 4 & -5 \end{bmatrix}$ & $B = \begin{bmatrix} 1 & 3 \\ -1 & 2 \end{bmatrix}$; find $|AB|$.

$$|AB| = |A||B| = (-6)(23) = -138$$

*** NOTE: Determinant of $\text{adj } A$

$|\text{adj } A| = |A|^{n-1}$ where n is order of matrix.

Eg: If $A = \begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix}$; find $|\text{adj } A|$.

$$|A| = 1(16-9) - 3(4-3) + 3(3-4) \\ = 7 - 3 - 3 = 1$$

$$|\text{adj } A| = |A|^{n-1} \\ = |1|^{3-1} = (1)^2 = 1$$

Eg: $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & -1 & 3 \end{bmatrix}$ $|A| = 1(6-3) - 1(3+6) + 1(-1-4) \\ = 3 - 9 - 5 = -11$

$$|\text{adj } A| = |A|^{n-1} \\ = (-11)^{3-1} = (-11)^2 = 121$$

Inverse of a Matrix:

By theorem, we know $A \cdot \text{adj } A = |A|I = \text{adj } A \cdot A \div |A|$

$$A \cdot \left(\frac{1}{|A|} \text{adj } A \right) = \left(\frac{1}{|A|} \text{adj } A \right) A = I$$

$$\Rightarrow A^{-1} = \frac{1}{|A|} \text{adj } A \quad (|A| \neq 0)$$

ie, A is non singular matrix.

Eg29] The sum of 3 no. is 6. If we multiply 3rd no. by 3 and add 2nd no. to it, we get 11. By adding 1st and 3rd nos. we get double the 2nd no. Find the no. using matrix method.

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Sol.] According to data the eq. are :

$$x + y + z = 6$$

$$3z + y = 11 \Rightarrow y + 3z = 11$$

$$x + z = 2y \Rightarrow x - 2y + z = 0$$

$$A X = B \rightarrow \textcircled{1}$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 3 \\ 1 & -2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 11 \\ 0 \end{bmatrix}$$

$$C_n = \begin{vmatrix} 1 & 3 & 0 & 3 & 0 & 1 \\ -2 & 1 & 1 & 1 & 1 & -2 \\ 1 & 1 & 1 & 1 & 1 & -2 \\ 1 & 3 & 0 & 3 & 0 & 1 \end{vmatrix} = \begin{bmatrix} 7 & 3 & -1 \\ -3 & 0 & 3 \\ 2 & -3 & 1 \end{bmatrix}$$

$$|A| = 7 + 3 - 1 = 9 ; \text{adj } A = C_n' = \begin{bmatrix} 7 & -3 & 2 \\ 3 & 0 & -3 \\ -1 & 3 & 1 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} \text{adj } A = \frac{1}{9} \begin{bmatrix} 7 & -3 & 2 \\ 3 & 0 & -3 \\ -1 & 3 & 1 \end{bmatrix}$$

$$\textcircled{1} \Rightarrow X = A^{-1} B$$

$$= \frac{1}{9} \begin{bmatrix} 7 & -3 & 2 \\ 3 & 0 & -3 \\ -1 & 3 & 1 \end{bmatrix} \begin{bmatrix} 6 \\ 11 \\ 0 \end{bmatrix}$$

$$= \frac{1}{9} \begin{bmatrix} 42 - 33 + 0 \\ 18 + 0 + 0 \\ -6 + 33 + 0 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{9} \begin{bmatrix} 9 \\ 18 \\ 27 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$\Rightarrow x = 1, y = 2, z = 3$$

4] Using cofactors of elements of 3rd column, evaluate

$$\Delta = \begin{vmatrix} 1 & x & yz \\ 1 & y & zx \\ 1 & z & xy \end{vmatrix}$$

$$\begin{aligned} \text{Along } C_3: \Delta &= yz \begin{vmatrix} 1 & y \\ 1 & z \end{vmatrix} - zx \begin{vmatrix} 1 & x \\ 1 & z \end{vmatrix} + xy \begin{vmatrix} 1 & x \\ 1 & y \end{vmatrix} \\ &= yz(z-y) - zx(z-x) + xy(y-x) \\ &= yz^2 - y^2z - z^2x + zx^2 + xy^2 - x^2y \end{aligned}$$

5] If $\Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$ and A_{ij} is cofactor of a_{ij} , then value of Δ is given by,

$$A) a_{11}A_{31} + a_{12}A_{32} + a_{13}A_{33} = 0$$

$$B) a_{11}A_{11} + a_{12}A_{21} + a_{13}A_{31} = 0$$

$$C) a_{21}A_{11} + a_{22}A_{12} + a_{23}A_{13} = 0$$

$$D) a_{11}A_{11} + a_{21}A_{21} + a_{31}A_{31} = \Delta$$

Properties of Determinants:

1] The value of Δ remains unchanged if its rows and columns are interchanged. (Transpose)

$$\begin{aligned} \text{Proof: } \Delta &= \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = a_1(b_2c_3 - b_3c_2) - a_2(b_1c_3 - b_3c_1) + a_3(b_1c_2 - b_2c_1) \\ &= a_1b_2c_3 - a_1b_3c_2 - a_2b_1c_3 + a_2b_3c_1 + a_3b_1c_2 - a_3b_2c_1 \end{aligned}$$

Rows $\leftrightarrow$ Columns

$$\begin{aligned} \Delta' &= \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = a_1(b_2c_3 - b_3c_2) - b_1(a_2c_3 - a_3c_2) + c_1(a_2b_3 - a_3b_2) \\ &= a_1b_2c_3 - a_1b_3c_2 - a_2b_1c_3 + a_2b_3c_1 + a_3b_1c_2 - a_3b_2c_1 \\ &= \Delta \end{aligned}$$

$$\therefore \Delta' = \Delta$$

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2] If any 2 rows (or columns) are interchanged in a Δ , then the sign of the Δ changes.

Eg: $A = \begin{vmatrix} 2 & 3 & 1 \\ -1 & 0 & 2 \\ 0 & 1 & -5 \end{vmatrix} = 2(0-2) - 3(5-0) + 1(-1-0)$
 $= -4 - 15 - 1 = -20$

$R_1 \leftrightarrow R_2$

$\Delta' = \begin{vmatrix} -1 & 0 & 2 \\ 2 & 3 & 1 \\ 0 & 1 & -5 \end{vmatrix} = -1(-15-1) - 0 + 2(2-0)$
 $= 16 + 4 = 20 = -\Delta$

Proof: $\Delta = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = a_1(b_2c_3 - b_3c_2) - a_2(b_1c_3 - b_3c_1) + a_3(b_1c_2 - b_2c_1)$
 $= a_1b_2c_3 - a_1b_3c_2 - a_2b_1c_3 + a_2b_3c_1 + a_3b_1c_2 - a_3b_2c_1$

$R_1 \leftrightarrow R_2$

$\Delta' = \begin{vmatrix} b_1 & b_2 & b_3 \\ a_1 & a_2 & a_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = b_1(a_2c_3 - a_3c_2) - b_2(a_1c_3 - a_3c_1) + b_3(a_1c_2 - a_2c_1)$
 $= a_2b_1c_3 - a_3b_1c_2 - a_1b_2c_3 + a_3b_2c_1 + a_1b_3c_2 - a_2b_3c_1$
 $= -(a_1b_2c_3 - a_1b_3c_2 - a_2b_1c_3 + a_2b_3c_1 + a_3b_1c_2 - a_3b_2c_1)$
 $\therefore \Delta' = -\Delta$

3] If any 2 rows (or 2 columns) of a Δ are identical, then value of Δ is zero.

Eg: $\Delta = \begin{vmatrix} 1 & 2 & 4 \\ 1 & 2 & 4 \\ 5 & 3 & 7 \end{vmatrix} = 1(14-20) + 2(7-20) + 4(3-10)$
 $= 2 + 26 - 28 = 0$

Proof: Let $R_1 \sim R_2$

$\Delta' = \begin{vmatrix} a_1 & a_2 & a_3 \\ a_1 & a_2 & a_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = a_1(a_2c_3 - a_3c_2) - a_2(a_1c_3 - a_3c_1) + a_3(a_1c_2 - a_2c_1)$
 $= a_1a_2c_3 - a_1a_3c_2 - a_1a_2c_3 + a_2a_3c_1 + a_1a_3c_2 - a_2a_3c_1$
 $= 0$

** 4] If Each element of a row (or a column) of a Δ is multiplied by a constant 'k', then its value gets multiplied by k.

Eg: $\Delta = \begin{vmatrix} 2 & 4 \\ 1 & 6 \end{vmatrix} = 12 - 4 = 8$

Let $R_1 \rightarrow 2R_1$

$$\Delta' = \begin{vmatrix} 4 & 8 \\ 1 & 6 \end{vmatrix} = 24 - 8 = 16 = 2(8)$$

$$\Delta' = 2\Delta$$

Proof: $\Delta = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = a_1(b_2c_3 - b_3c_2) - a_2(b_1c_3 - b_3c_1) + a_3(b_1c_2 - b_2c_1)$

Let $R_1 \rightarrow kR_1$

$$\Delta' = \begin{vmatrix} ka_1 & ka_2 & ka_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = ka_1(b_2c_3 - b_3c_2) - ka_2(b_1c_3 - b_3c_1) + ka_3(b_1c_2 - b_2c_1)$$

$$= k[a_1(b_2c_3 - b_3c_2) - a_2(b_1c_3 - b_3c_1) + a_3(b_1c_2 - b_2c_1)]$$

$$= k\Delta$$

$$\therefore \Delta' = k\Delta$$

** 5] If some or all elements of a row (or column) of a Δ are expressed as sum of 2 (or more) terms, then the Δ can be expressed as sum of 2 (or more) Δ 's.

Eg 10] Show that $\begin{vmatrix} a & b & c \\ a+2x & b+2y & c+2z \\ x & y & z \end{vmatrix} = 0$

$$\text{LHS} = \begin{vmatrix} a & b & c \\ a+2x & b+2y & c+2z \\ x & y & z \end{vmatrix} = \begin{vmatrix} a & b & c \\ a & b & c \\ x & y & z \end{vmatrix} + \begin{vmatrix} 2x & 2y & 2z \\ a & b & c \\ x & y & z \end{vmatrix}$$

$$\because R_1 \sim R_2$$

$$= 0 + 2 \begin{vmatrix} a & b & c \\ x & y & z \\ x & y & z \end{vmatrix}$$

$$\because R_2 \sim R_3$$

$$= 0 + 2(0)$$

$$= 0 = \text{RHS}$$

Proof: Let elements of R_1 be added with $\lambda_1, \lambda_2, \lambda_3$ respectively.

$$\Delta' = \begin{vmatrix} a_1 + \lambda_1 & a_2 + \lambda_2 & a_3 + \lambda_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

$$= (a_1 + \lambda_1)(b_2 c_3 - b_3 c_2) - (a_2 + \lambda_2)(b_1 c_3 - b_3 c_1) + (a_3 + \lambda_3)(b_1 c_2 - b_2 c_1)$$

$$= [a_1(b_2 c_3 - b_3 c_2) - a_2(b_1 c_3 - b_3 c_1) + a_3(b_1 c_2 - b_2 c_1)] + [\lambda_1(b_2 c_3 - b_3 c_2) - \lambda_2(b_1 c_3 - b_3 c_1) + \lambda_3(b_1 c_2 - b_2 c_1)]$$

$$= \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} + \begin{vmatrix} \lambda_1 & \lambda_2 & \lambda_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

Hence, proved.

- 6) If, to each element of any row (or column) of a Δ , the equimultiples of corresponding elements of other row or column are added, then value of Δ remains the same, i.e., $R_i \rightarrow R_i + kR_j$ or $C_i \rightarrow C_i + kC_j$.

Proof:

$$\Delta = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

$$R_1 \rightarrow R_1 + kR_3$$

$$\Delta' = \begin{vmatrix} a_1 + kc_1 & a_2 + kc_2 & a_3 + kc_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

$$= \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} + \begin{vmatrix} kc_1 & kc_2 & kc_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

$$\Delta' = \Delta + k \begin{vmatrix} c_1 & c_2 & c_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

$$\Delta' = \Delta + 0 \quad (\because R_1 \sim R_3)$$

$$\Rightarrow \Delta' = \Delta$$

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7] Value of the triangular Δ is product of the PD elements.

$$\text{Proof: } \Delta' = \begin{vmatrix} a_1 & a_2 & a_3 \\ 0 & b_2 & b_3 \\ 0 & 0 & c_3 \end{vmatrix} = a_1(b_2c_3 - 0) - a_2(0-0) + a_3(0-0)$$

$$= a_1b_2c_3 - 0 - 0 - 0 - 0$$

$$= a_1b_2c_3$$

8] If all elements of any one row (or column) are equal to zero, value of the Δ is zero.

$$\text{Proof: } \Delta' = \begin{vmatrix} a_1 & a_2 & a_3 \\ 0 & 0 & 0 \\ c_1 & c_2 & c_3 \end{vmatrix} = a_1(0-0) - a_2(0-0) + a_3(0-0)$$

$$= 0 - 0 + 0$$

$$= 0$$

EXERCISE 4.2

1] Prove that, $\begin{vmatrix} x & a & x+a \\ y & b & y+b \\ z & c & z+c \end{vmatrix} = 0$

$$C_1 \rightarrow C_1 + C_2$$

$$\text{LHS} = \begin{vmatrix} x+a & a & x+a \\ y+b & b & y+b \\ z+c & c & z+c \end{vmatrix} = 0 \quad (\because C_1 \sim C_3)$$

2] Prove that, $\begin{vmatrix} a-b & b-c & c-a \\ b-c & c-a & a-b \\ c-a & a-b & b-c \end{vmatrix} = 0$

$$C_1 \rightarrow C_1 + C_2 + C_3$$

$$\begin{vmatrix} 0 & b-c & c-a \\ 0 & c-a & a-b \\ 0 & a-b & b-c \end{vmatrix} = 0 \quad (\because C_1 = 0)$$

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** 3] Prove that $\begin{vmatrix} 2 & 7 & 65 \\ 3 & 8 & 75 \\ 5 & 9 & 86 \end{vmatrix} = 0$

$$\text{LHS} = \begin{vmatrix} 2 & 7 & 65 \\ 3 & 8 & 75 \\ 5 & 9 & 86 \end{vmatrix}$$

$$C_2 \rightarrow 9C_2 + C_1$$

$$= \begin{vmatrix} 2 & 65 & 65 \\ 3 & 75 & 75 \\ 5 & 86 & 86 \end{vmatrix} = 0 \quad (\because C_2 \sim C_3)$$

4] Prove that $\begin{vmatrix} 1 & bc & a(b+c) \\ 1 & ca & b(c+a) \\ 1 & ab & c(a+b) \end{vmatrix} = 0$

$$\text{LHS} = \begin{vmatrix} 1 & bc & a(b+c) \\ 1 & ca & b(c+a) \\ 1 & ab & c(a+b) \end{vmatrix}$$

$$C_3 \rightarrow C_3 + C_2$$

$$= \begin{vmatrix} 1 & bc & ab+bc+ca \\ 1 & ca & ab+bc+ca \\ 1 & ab & ab+bc+ca \end{vmatrix} = (ab+bc+ca) \begin{vmatrix} 1 & bc & 1 \\ 1 & ca & 1 \\ 1 & ab & 1 \end{vmatrix}$$

$$= (ab+bc+ca)(0) = 0 \quad (\because C_1 \sim C_3)$$

5] Prove that $\begin{vmatrix} b+c & q+r & y+z \\ c+a & r+p & z+x \\ a+b & p+q & x+y \end{vmatrix} = 2 \begin{vmatrix} a & p & x \\ b & q & y \\ c & r & z \end{vmatrix}$

$$\text{LHS} = \begin{vmatrix} b+c & q+r & y+z \\ c+a & r+p & z+x \\ a+b & p+q & x+y \end{vmatrix}$$

$$R_1 \rightarrow R_1 + R_2 + R_3$$

$$= \begin{vmatrix} 2(a+b+c) & 2(p+q+r) & 2(x+y+z) \\ c+a & r+p & z+x \\ a+b & p+q & x+y \end{vmatrix}$$

$$= 2 \begin{vmatrix} a+b+c & p+q+r & x+y+z \\ c+a & r+p & z+x \\ a+b & p+q & x+y \end{vmatrix}$$

$$R_2 \rightarrow R_2 - R_1; R_3 \rightarrow R_3 - R_1$$

$$= 2 \begin{vmatrix} a+b+c & p+q+r & x+y+z \\ -b & -q & -y \\ -c & -r & -z \end{vmatrix}$$

$$R_1 \rightarrow R_1 + R_2 + R_3$$

$$= 2 \begin{vmatrix} a & p & x \\ -b & -q & -y \\ -c & -r & -z \end{vmatrix} = (-1)(-1)(-1) 2 \begin{vmatrix} a & p & x \\ b & q & y \\ c & r & z \end{vmatrix} = 2 \begin{vmatrix} a & p & x \\ b & q & y \\ c & r & z \end{vmatrix}$$

6] $\begin{vmatrix} 0 & a & -b \\ -a & 0 & -c \\ b & c & 0 \end{vmatrix} = 0$: Prove

3rd ordered skew symmetric matrix
 $\therefore \Delta = 0$

7] Prove that $\begin{vmatrix} -a^2 & ab & ac \\ ba & -b^2 & bc \\ ca & cb & -c^2 \end{vmatrix} = 4a^2b^2c^2$

LHS = $\begin{vmatrix} -a^2 & ab & ac \\ ba & -b^2 & bc \\ ca & cb & -c^2 \end{vmatrix}$ Take out a, b, c common from R_1, R_2, R_3 respectively.

$$= abc \begin{vmatrix} -a & b & c \\ a & -b & c \\ a & b & -c \end{vmatrix}$$

$$= a^2b^2c^2 \begin{vmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{vmatrix}$$

$$\begin{aligned}
 &= a^2 b^2 c^2 [-1(1-1) - 1(-1-1) + 1(1+1)] \\
 &= a^2 b^2 c^2 [0 + 2 + 2] \\
 &= 4a^2 b^2 c^2
 \end{aligned}$$

Type I

$$8] (a) \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} = (a-b)(b-c)(c-a)$$

$$(b) \begin{vmatrix} 1 & a & a^3 \\ 1 & b & b^3 \\ 1 & c & c^3 \end{vmatrix} = (a-b)(b-c)(c-a)(a+b+c)$$

$$(c) \begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix} = (a-b)(b-c)(c-a)(ab+bc+ca)$$

$$(a) \text{ LHS} = \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} \quad R_2 \rightarrow R_2 - R_1; \quad R_3 \rightarrow R_3 - R_1$$

$$= \begin{vmatrix} 1 & a & a^2 \\ 0 & b-a & b^2-a^2 \\ 0 & c-a & c^2-a^2 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & a & a^2 \\ 0 & b-a & (b-a)(b+a) \\ 0 & c-a & (c-a)(c+a) \end{vmatrix} = 2H1 (9)$$

$$= (b-a)(c-a) \begin{vmatrix} 1 & a & a^2 \\ 0 & 1 & b+a \\ 0 & 0 & c+a \end{vmatrix}$$

$$R_3 \rightarrow R_3 - R_2$$

$$= (b-a)(c-a) \begin{vmatrix} 1 & a & a^2 \\ 0 & 1 & b+a \\ 0 & 0 & c-b \end{vmatrix}$$

$$\begin{aligned}
 &= -(a-b)(c-a)(c-b) \\
 &= (a-b)(b-c)(c-a) \\
 &= \text{RHS}
 \end{aligned}$$

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(b) LHS = $\begin{vmatrix} 1 & a & a^3 \\ 1 & b & b^3 \\ 1 & c & c^3 \end{vmatrix}$ $R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_3 - R_1$

$= \begin{vmatrix} 1 & a & a^3 \\ 0 & b-a & b^3-a^3 \\ 0 & c-a & c^3-a^3 \end{vmatrix}$ $a^3-b^3 = (a-b)(a^2+ab+b^2)$

$= \begin{vmatrix} 1 & a & a^3 \\ 0 & b-a & (b-a)(b^2+ab+a^2) \\ 0 & c-a & (c-a)(c^2+ac+a^2) \end{vmatrix}$

$= (b-a)(c-a) \begin{vmatrix} 1 & a & a^3 \\ 0 & 1 & a^2+ab+b^2 \\ 0 & 1 & c^2+ac+a^2 \end{vmatrix}$

$R_3 \rightarrow R_3 - R_2: = -(a-b)(c-a) \begin{vmatrix} 1 & a & a^3 \\ 0 & 1 & a^2+ab+b^2 \\ 0 & 0 & (c-b)(c+b)+a(c-b) \end{vmatrix}$

$= -(a-b)(c-a)(c-b) \begin{vmatrix} 1 & a & a^3 \\ 0 & 1 & a^2+ab+b^2 \\ 0 & 0 & a+b+c \end{vmatrix}$

$= (a-b)(b-c)(c-a)(a+b+c)$

(c) LHS = $\begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix}$ $R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_3 - R_1$

$= \begin{vmatrix} 1 & a^2 & a^3 \\ 0 & b^2-a^2 & b^3-a^3 \\ 0 & c^2-a^2 & c^3-a^3 \end{vmatrix}$

$= \begin{vmatrix} 1 & a^2 & a^3 \\ 0 & (b-a)(b+a) & (b-a)(b^2+ab+a^2) \\ 0 & (c-a)(c+a) & (c-a)(c^2+ac+a^2) \end{vmatrix}$

$$\begin{aligned}
 &= (b-a)(c-a) \begin{vmatrix} 1 & a^2 & a^3 \\ 0 & b+a & b^2+ab+a^2 \\ 0 & c+a & c^2+ab+a^2 \end{vmatrix} \\
 &R_3 \rightarrow R_3 - R_2 \\
 &= -(a-b)(c-a) \begin{vmatrix} 1 & a^2 & a^3 \\ 0 & b+a & b^2+ab+a^2 \\ 0 & c-b & (c-b)(c+b)+a(c-b) \end{vmatrix} \\
 &= -(a-b)(c-a)(c-b) \begin{vmatrix} 1 & a^2 & a^3 \\ 0 & b+a & a^2+ab+b^2 \\ 0 & 1 & a+b+c \end{vmatrix} \\
 &= (a-b)(b-c)(c-a) [1(ab+b^2+bc+a^2+ab+ac-b^2-ab-a^2)] \\
 &= (a-b)(b-c)(c-a)(ab+bc+ca)
 \end{aligned}$$

9) $\begin{vmatrix} x & x^2 & yz \\ y & y^2 & zx \\ z & z^2 & xy \end{vmatrix} = (x-y)(y-z)(z-x)(xy+yz+zx)$

LHS = $\begin{vmatrix} x & x^2 & yz \\ y & y^2 & zx \\ z & z^2 & xy \end{vmatrix}$

$R_1 \rightarrow R_1 - R_2; R_2 \rightarrow R_2 - R_3$

$$= \begin{vmatrix} x-y & x^2-y^2 & yz-zx \\ y-z & y^2-z^2 & -x(y-z) \\ z & z^2 & xy \end{vmatrix} = \begin{vmatrix} x-y & (x-y)(x+y) & -z(x-y) \\ y-z & (y-z)(y+z) & -x(y-z) \\ z & z^2 & xy \end{vmatrix}$$

$$= (x-y)(y-z) \begin{vmatrix} 1 & x+y & -z \\ 1 & y+z & -x \\ z & z^2 & xy \end{vmatrix}$$

$R_1 \rightarrow R_1 - R_2$

$$= (x-y)(y-z) \begin{vmatrix} 0 & (x-z) & (x-z) \\ 1 & y+z & -x \\ z & z^2 & xy \end{vmatrix}$$

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$$= (x-y)(y-z)(z-x) \begin{vmatrix} 0 & 1 & 1 \\ 1 & y+z & -x \\ z & z^2 & xy \end{vmatrix}$$

$$= (x-y)(y-z)(z-x) [-1(xy+xz) + 1(z^2-zy-z^2)]$$

$$= (x-y)(y-z)(z-x)(-xy-xz-zy)$$

$$= (x-y)(y-z)(z-x)(xy+xz+zy)$$

Type II
10] i) $(R_1+R_2+R_3)_{on}$
 $(C_1+C_2+C_3)$

$$\begin{vmatrix} x+4 & 2x & 2x \\ 2x & x+4 & 2x \\ 2x & 2x & x+4 \end{vmatrix} = (5x+4)(4-x)^2$$

LHS $= R_1 \rightarrow R_1 + R_2 + R_3$

$$= \begin{vmatrix} 5x+4 & 5x+4 & 5x+4 \\ 2x & x+4 & 2x \\ 2x & 2x & x+4 \end{vmatrix} = (5x+4) \begin{vmatrix} 1 & 1 & 1 \\ 2x & x+4 & 2x \\ 2x & 2x & x+4 \end{vmatrix}$$

$C_2 \rightarrow C_2 - C_1$; $C_3 \rightarrow C_3 - C_1$

$$= (5x+4) \begin{vmatrix} 1 & 0 & 0 \\ 2x & 4-x & 0 \\ 2x & 0 & 4-x \end{vmatrix}$$

$$= (5x+4)(4-x)^2 = \text{RHS}$$

ii) $\begin{vmatrix} y+k & y & y \\ y & y+k & y \\ y & y & y+k \end{vmatrix} = (3y+k)(k^2)$

LHS $= C_1 \rightarrow C_1 + C_2 + C_3$

$$= \begin{vmatrix} 3y+k & y & y \\ 3y+k & y+k & y \\ 3y+k & y & y+k \end{vmatrix} = (3y+k) \begin{vmatrix} 1 & y & y \\ 1 & y+k & y \\ 1 & y & y+k \end{vmatrix}$$

$R_2 \rightarrow R_2 - R_1$; $R_3 \rightarrow R_3 - R_1$

$$= (3y+k) \begin{vmatrix} 1 & y & y \\ 0 & k & 0 \\ 0 & 0 & k \end{vmatrix}$$

$$= (3y+k)(k^2) = \text{RHS}$$

$$\begin{aligned} ({}^s d + {}^s a + 1) \cdot 0 &= ({}^s d + {}^s a + 1) \cdot ({}^s d + {}^s a + 1) = \\ &({}^s d + {}^s a + 1) \cdot ({}^s d + {}^s a + 1) = \\ &({}^s d + {}^s a + 1) \cdot ({}^s d + {}^s a + 1) = \end{aligned}$$

$$R_2 \rightarrow R_2 - R_1 ; R_3 \rightarrow R_3 - R_1$$

$$= \begin{vmatrix} 1 & x & y \\ 0 & y+z+x & 0 \\ 0 & 0 & z+x+y \end{vmatrix}$$

$$= 2(x+y+z)(x+y+z)^2$$

$$= 2(x+y+z)^3$$

*** (12)

$$\begin{vmatrix} 1 & x & x^2 \\ x^2 & 1 & x \\ x & x^2 & 1 \end{vmatrix} = (1-x^3)^2$$

$$\text{LHS} = \begin{vmatrix} 1 & x & x^2 \\ x^2 & 1 & x \\ x & x^2 & 1 \end{vmatrix}$$

$$R_1 \rightarrow R_1 - xR_2 ; R_2 \rightarrow R_2 - xR_3$$

$$= \begin{vmatrix} 1-x^3 & 0 & 0 \\ 0 & 1-x^3 & 0 \\ x & x^2 & 1 \end{vmatrix} = (1-x^3)^2$$

*** (13)

$$\text{Prove that } \begin{vmatrix} 1+a^2-b^2 & 2ab & -2b \\ 2ab & 1-a^2+b^2 & 2a \\ 2b & -2a & 1-a^2-b^2 \end{vmatrix} = (1+a^2+b^2)^3$$

$$\text{LHS: } C_1 \rightarrow C_1 - bC_3 ; C_2 \rightarrow C_2 + aC_3$$

$$= \begin{vmatrix} 1+a^2+b^2 & 0 & -2b \\ 0 & 1+a^2+b^2 & 2a \\ b(1+a^2+b^2) & -a(1+a^2+b^2) & 1-a^2-b^2 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 0 & -2b \\ (1+a^2+b^2)^2 & 0 & 2a \\ b & -a & 1-a^2-b^2 \end{vmatrix}$$

$$= (1+a^2+b^2)^2 (1-a^2-b^2+2a^2+2b^2)$$

$$= (1+a^2+b^2)^2 (1+a^2+b^2)$$

$$= (1+a^2+b^2)^3$$

$$^{**} 14] \begin{vmatrix} a^2+1 & ab & ac \\ ab & b^2+1 & bc \\ ca & cb & c^2+1 \end{vmatrix} = 1+a^2+b^2+c^2$$

LHS : Multiply and divide R_1, R_2, R_3 by a, b, c respectively.

$$= \frac{1}{abc} \begin{vmatrix} a(a^2+1) & a^2b & a^2c \\ ab^2 & b(b^2+1) & b^2c \\ ac^2 & bc^2 & c(c^2+1) \end{vmatrix}$$

Take out a, b, c common from C_1, C_2, C_3 respectively.

$$= \frac{abc}{abc} \begin{vmatrix} 1+a^2 & a^2 & a^2 \\ b^2 & b^2+1 & b^2 \\ c^2 & c^2 & c^2+1 \end{vmatrix}$$

$$R_1 \rightarrow R_1 + R_2 + R_3$$

$$= \begin{vmatrix} 1+a^2+b^2+c^2 & 1+a^2+b^2+c^2 & 1+a^2+b^2+c^2 \\ b^2 & b^2+1 & b^2 \\ c^2 & c^2 & c^2+1 \end{vmatrix}$$

$$= 1+a^2+b^2+c^2 \begin{vmatrix} 1 & 1 & 1 \\ b^2 & b^2+1 & b^2 \\ c^2 & c^2 & c^2+1 \end{vmatrix}$$

$$C_2 \rightarrow C_2 - C_1; \quad C_3 \rightarrow C_3 - C_1$$

$$= 1+a^2+b^2+c^2 \begin{vmatrix} 1 & 0 & 0 \\ b^2 & 1 & 0 \\ c^2 & 0 & 1 \end{vmatrix}$$

$$= (1+a^2+b^2+c^2)(1)$$

$$= 1+a^2+b^2+c^2$$

$$= \text{RHS}$$

Eg12] Without expanding prove that $\begin{vmatrix} x+y & y+z & z+x \\ z & 1+x & y \\ 1 & y & 1+x \end{vmatrix} = 0$

$$R_1 \rightarrow R_1 + R_2$$

$$\text{LHS: } \begin{vmatrix} x+y+z & x+y+z & x+y+z \\ z & 1+x & y \\ 1 & y & 1+x \end{vmatrix}$$

$$= x+y+z \begin{vmatrix} 1 & 1 & 1 \\ z & 1+x & y \\ 1 & y & 1+x \end{vmatrix}$$

$$= 0 \quad [\because R_1 \sim R_3]$$

Eg11] $\begin{vmatrix} a & a+b & a+b+c \\ 2a & 3a+2b & 4a+3b+2c \\ 3a & 6a+3b & 10a+6b+3c \end{vmatrix} = a^3$

$$R_2 \rightarrow R_2 - 2R_1; R_3 \rightarrow R_3 - 3R_1$$

$$\text{LHS: } \begin{vmatrix} a & a+b & a+b+c \\ 0 & a & 2a+b \\ 0 & 3a & 7a+3b \end{vmatrix}$$

$$= \begin{vmatrix} a & a+b & a+b+c \\ 0 & a & 2a+b \\ 0 & 0 & a \end{vmatrix}$$

$$= a^3 = \text{RHS}$$

Eg13] Evaluate: $\begin{vmatrix} 1 & a & bc \\ 1 & b & ca \\ 1 & c & ab \end{vmatrix}$

$$R_2 \rightarrow R_2 - R_1; R_3 \rightarrow R_3 - R_1$$

$$\begin{aligned}
 \text{LHS: } \Delta &= \begin{vmatrix} 1 & a & bc \\ 0 & b-a & ca-bc \\ 0 & c-a & ab-bc \end{vmatrix} \\
 &= \begin{vmatrix} 1 & a & bc \\ 0 & b-a & -c(b-a) \\ 0 & c-a & -b(c-a) \end{vmatrix} \\
 &= (b-a)(c-a) \begin{vmatrix} 1 & a & b \\ 0 & 1 & -c \\ 0 & 1 & -b \end{vmatrix} \\
 &= (b-a)(c-a) \begin{vmatrix} 1 & a & b \\ 0 & 1 & -c \\ 0 & 0 & -b+c \end{vmatrix} \quad (R_3 \rightarrow R_3 - R_2) \\
 &= (b-a)(c-a)(c-b) \\
 &= (-1)(a-b)(-1)(b-c)(c-a) \\
 &= (a-b)(b-c)(c-a)
 \end{aligned}$$

Eg 14] $\begin{vmatrix} b+c & a & a \\ b & c+a & b \\ c & c & a+b \end{vmatrix} = 4abc$

$$\begin{aligned}
 \text{LHS: } \Delta &= \begin{vmatrix} b+c & a & a \\ b & c+a & b \\ c & c & a+b \end{vmatrix} \\
 &= \begin{vmatrix} 0 & -2c & -2b \\ b & c+a & b \\ c & c & a+b \end{vmatrix} \quad (R_1 \rightarrow R_1 - R_2 - R_3) \\
 &= 2 \begin{vmatrix} 0 & -c & -b \\ b & c+a & b \\ c & c & a+b \end{vmatrix} \\
 &= 2 \begin{vmatrix} 0 & -c & -b \\ b & a & 0 \\ c & 0 & a \end{vmatrix} \quad (R_2 \rightarrow R_2 + R_1; R_3 \rightarrow R_3 + R_1) \\
 &= 2 [c(ab) - b(-ac)] \\
 &= 2(abc + abc) \\
 &= 2(2abc) = 4abc
 \end{aligned}$$

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Eg[5] If x, y, z are all different and $\Delta = \begin{vmatrix} x & x^2 & 1+x^3 \\ y & y^2 & 1+y^3 \\ z & z^2 & 1+z^3 \end{vmatrix} = 0$

then show that $1+xyz = 0$

$\Delta \Rightarrow \begin{vmatrix} x & x^2 & 1+x^3 \\ y & y^2 & 1+y^3 \\ z & z^2 & 1+z^3 \end{vmatrix} = 0$

$\Rightarrow \begin{vmatrix} x & x^2 & 1 \\ y & y^2 & 1 \\ z & z^2 & 1 \end{vmatrix} + \begin{vmatrix} x & x^2 & x^3 \\ y & y^2 & y^3 \\ z & z^2 & z^3 \end{vmatrix} = 0$

$C_1 \leftrightarrow C_3 ; C_2 \leftrightarrow C_3$

$\Rightarrow (-)(-) \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} + xyz \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} = 0$

$\Rightarrow \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} (1+xyz) = 0$

$\therefore R_2 \rightarrow R_2 - R_1 ; R_3 \rightarrow R_3 - R_1$

$\Rightarrow \begin{vmatrix} 1 & x & x^2 \\ 0 & y-x & y^2-x^2 \\ 0 & z-x & z^2-x^2 \end{vmatrix} (1+xyz) = 0$

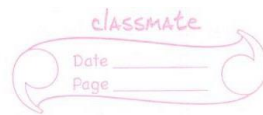
$\Rightarrow \begin{vmatrix} 1 & x & x^2 \\ 0 & y-x & (y-x)(y+x) \\ 0 & z-x & (z-x)(z+x) \end{vmatrix} (1+xyz) = 0$

$(y-x)(z-x) \begin{vmatrix} 1 & x & x^2 \\ 0 & 1 & y+x \\ 0 & 1 & z+x \end{vmatrix} (1+xyz) = 0$

$(y-x)(z-x) \begin{vmatrix} 1 & x & x^2 \\ 0 & 1 & y+x \\ 0 & 0 & z-y \end{vmatrix} (1+xyz) = 0$

$(y-x)(z-x)(z-y)(1+xyz) = 0$

$\therefore 1+xyz = 0$



Eg 16] Show that $\begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix} = abc \left(1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right)$

$$\text{LHS} = \begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix}$$

Take out a, b, c common from R_1, R_2, R_3 respectively.

$$= abc \begin{vmatrix} \frac{1}{a}+1 & \frac{1}{a} & \frac{1}{a} \\ \frac{1}{b} & \frac{1}{b}+1 & \frac{1}{b} \\ \frac{1}{c} & \frac{1}{c} & \frac{1}{c}+1 \end{vmatrix}$$

$$R_1 \rightarrow R_1 + R_2 + R_3$$

$$= abc \begin{vmatrix} 1+\frac{1}{a}+\frac{1}{b}+\frac{1}{c} & 1+\frac{1}{a}+\frac{1}{b}+\frac{1}{c} & 1+\frac{1}{a}+\frac{1}{b}+\frac{1}{c} \\ \frac{1}{b} & \frac{1}{b}+1 & \frac{1}{b} \\ \frac{1}{c} & \frac{1}{c} & \frac{1}{c}+1 \end{vmatrix}$$

$$= abc \left(1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) \begin{vmatrix} 1 & 1 & 1 \\ \frac{1}{b} & \frac{1}{b}+1 & \frac{1}{b} \\ \frac{1}{c} & \frac{1}{c} & \frac{1}{c}+1 \end{vmatrix}$$

$$C_2 \rightarrow C_2 - C_1, \quad C_3 \rightarrow C_3 - C_1$$

$$= (abc) \left(1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) \begin{vmatrix} 1 & 0 & 0 \\ \frac{1}{b} & 1 & 0 \\ \frac{1}{c} & 0 & 1 \end{vmatrix}$$

$$= (abc) \left(1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) (1)$$

$$= (abc) \left(1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right)$$

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$$\Rightarrow (y-x)(z-x) \begin{vmatrix} 1 & x & x^2 \\ 0 & 1 & y+x \\ 0 & 1 & z+x \end{vmatrix} (1+xyz) = 0$$

$$\Rightarrow (-)(x-y)(z-x) R_3 \rightarrow R_3 - R_2$$

$$\Rightarrow (-)(x-y)(z-x) \begin{vmatrix} 1 & x & x^2 \\ 0 & 1 & y+x \\ 0 & 0 & z-y \end{vmatrix} (1+xyz) = 0$$

$$\Rightarrow (-)(x-y)(z-x)(z-y)(1+xyz) = 0$$

$$\Rightarrow (-)(-)(x-y)(y-z)(z-x)(1+xyz) = 0$$

$$\Rightarrow (x-y)(y-z)(z-x)(1+xyz) = 0$$

Given x, y, z are all different
 $\Rightarrow$ only $(1+xyz) = 0$